

# Strategic Responses to Portfolio Reallocation: Evidence from Dividends\*

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## Abstract

To attract investor capital, firms adjust dividends to satisfy investors' preferences. Yet investors rarely reallocate their capital. When, then, do firms adjust dividends? This paper shows that firms strategically increase dividends when shareholders reconsider their portfolios. I identify such periods using large mutual fund flows, which are associated with more discretionary trading. Firms exposed to large flows significantly increase dividends. The response is symmetric across inflows and outflows, and announcements follow in the quarter immediately after large flows, consistent with firms timing payout to windows of reallocation. A difference-in-differences design exploiting the introduction of Morningstar Sustainability Ratings supports causality. The effect is stronger among firms with high income fund ownership, consistent with catering. Funds subsequently favor firms that raise dividends following both inflow and outflow shocks. These findings highlight a strategic interaction between firms and their investors around periods of portfolio reallocation.

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# 1 Introduction

Firms care about investor demand for their shares. For example, they voluntarily adopt International Accounting Standards to appeal to foreign investors (Covrig et al., 2007) and invest in investor relations activities to attract institutional investors (Bushee and Miller, 2012). Dividend policy can be used for a similar purpose. Firms adjust dividends to align with investors’ dividend preferences (Allen et al., 2000; Baker and Wurgler, 2004; Golubov et al., 2020).

Yet growing evidence suggests that investor capital is not easily moved. Firm characteristics, including dividends, have limited power in explaining stock demand (Koijen and Yogo, 2019), investors slowly incorporate information into their portfolios because of limited attention (Cohen and Frazzini, 2008; Schmidt, 2019), and investors tend to maintain their existing portfolio weights (Brunnermeier and Nagel, 2008; Lou, 2012). While the dividend catering literature assumes that investors move capital toward firms whose dividend policies match their preferences, this evidence suggests that capital does not always follow. Firms may therefore strategically choose not only how much to pay in dividends but also when to pay them. This paper asks whether firms increase dividends when their shareholders are actively reallocating their portfolios.

To answer this question, I use large mutual fund flows to identify periods in which fund managers are likely to reconsider their portfolios. I first examine whether large flows indeed induce active portfolio rebalancing by fund managers. I then test whether firms strategically increase dividends when fund managers are forced to reconsider their portfolios, and whether these dividend increases successfully attract subsequent mutual fund investment. My analysis combines international data on equity mutual funds, institutional portfolio holdings, and publicly listed firms, constructing a sample of 14,593 funds and 17,853 firms over the 2000–2021 period.

Large fund flows are well suited to identifying periods of portfolio reassessment for two reasons. First, flows are determined at the fund level rather than at the level of a particular holding, mitigating concerns that they directly reflect firm-specific information. Second, and more importantly, while typical flows can be absorbed by proportional scaling, large flows cannot be. As liquidity and price impact constraints bind, managers need to decide which positions to adjust and by how much. Practitioners also argue that sizable inflows or outflows are difficult to accommodate within existing portfolio strategies and may force fund managers to revisit those strategies.<sup>1</sup> Thus, large flows help identify periods in which deliberate portfolio decisions are more likely to occur.

I first verify this premise by examining how fund managers adjust their portfolios

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<sup>1</sup>Morningstar (2015) quotes an industry veteran at a major asset management group: “Heavy inflows or outflows have a negative impact on investment strategy implementation.” Morningstar (2017) further reports that large inflows or outflows can be very disruptive, noting that they absorb a substantial portion of fund managers’ time and may necessitate adjustments in portfolio management strategies.

in response to large fund flows. I find a strong positive association between the size of fund flows and discretionary trading. Quantitatively, a one-standard-deviation increase in absolute flows is associated with a 32% rise in deviations from proportional flow allocation, and this pattern holds in both inflow and outflow periods. Rather than simply scaling positions proportionally, fund managers make more active and deliberate trading decisions when facing larger flows.

Building on this evidence, I next examine whether firms increase dividends when fund flows prompt fund managers to reconsider their portfolios. Firms exposed to larger absolute flows increase dividends: a one-standard-deviation increase in flow exposure corresponds to a 1.5% rise in dividends relative to the unconditional mean. Importantly, dividends increase following both large inflows and outflows. The symmetric response is consistent with firms responding to the underlying portfolio reallocation induced by large flows, rather than to the direction of trading itself.<sup>2</sup>

Two additional patterns further support the interpretation that dividends respond to portfolio reallocation. First, the dividend response is concentrated among firms exposed to extreme flows. This evidence is consistent with firms responding when flows are large enough to trigger active portfolio reallocation. Second, the timing of the response is consistent with a deliberate adjustment of dividends by firms. Among firms that raise dividends, 26.8% of dividend announcements follow a quarter in the top quintile of flow-induced trading pressure, compared with 15.2% for the bottom quintile. Within the same firm-year, exposure to large flows in the immediately preceding quarter predicts subsequent announcements of dividend increases, whereas exposure in more distant quarters does not. These results suggest that firms strategically time their dividend increases around periods when their shareholders reconsider their portfolios.

While the results are consistent with strategic responses to active portfolio reallocation, they do not establish a causal effect. Although the use of fund flows mitigates concerns that the measure directly reflects firm-specific information, fund flows are related to fund performance and may thus be correlated with the characteristics of firms held by the fund. To address this concern, I exploit the 2016 introduction of Morningstar Sustainability Ratings as a source of exogenous variation in fund flows. Hartzmark and Sussman (2019) show that this rating publication generated significant inflows to funds receiving “Five Globes” and outflows from those receiving “One Globe” without affecting fund performance. I find similar flow patterns in my sample.

Leveraging this identification strategy, I implement a difference-in-differences (DiD) analysis by defining treated firms as those in the top quartile of ownership by either five-globe or one-globe funds as of December 2015. I find that treated firms increase their dividends following the flow shock, and I find no evidence of differential pre-trends.

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<sup>2</sup>In contrast, repurchases respond only to price pressure induced by outflows, consistent with a market timing motive.

Consistent with the baseline findings, dividend increases occur following both the inflow shock from five-globe funds and the outflow shock from one-globe funds, confirming the symmetric response. Collectively, these results strengthen the evidence that firms strategically increase dividends in response to portfolio reallocation and provide support for a causal interpretation.

Next, I test whether these strategic dividend increases are driven by a catering motive. If so, the response should be stronger among firms whose shareholders strongly prefer dividends. Consistent with this prediction, the response is stronger among firms with greater ownership by income funds, by funds domiciled in countries that cut dividend taxes, and by funds domiciled in countries with older populations.

Finally, I examine whether this strategic dividend policy generates a favorable investment response. Among treated firms, those that increase dividends are subsequently favored both by funds experiencing inflow shocks and by funds experiencing outflow shocks. This pattern suggests that dividend increases during portfolio reallocation can be effective not only in attracting new capital from funds experiencing inflows, but also in retaining investment from funds experiencing outflows.

I further verify the robustness of the dividend catering channel by ruling out confounding channels. In particular, I test whether a governance channel explains the results. Large flows may strengthen governance through intensified monitoring and the threat of exit during active rebalancing. Dividends can serve as a commitment device that disciplines corporate managers by reducing the free cash flow available for value-destroying investment (Easterbrook, 1984; Jensen, 1986). If governance drives the response, dividend increases should be concentrated among firms with greater potential for free cash flow problems. The results provide little evidence of the governance channel.

By studying dividend decisions when shareholders reconsider their holdings, this paper contributes to four strands of literature: dividend catering, the real effects of fund flows, flow-induced trading, and investor attention. First, this paper contributes to the dividend catering literature by showing when catering occurs. The existing literature has focused on establishing the existence of dividend catering, using the dividend premium (Baker and Wurgler, 2004; Li and Lie, 2006), tax regime changes (Chetty and Saez, 2005; Hanlon and Hoopes, 2014), and shareholder characteristics (Grinstein and Michaely, 2005; Becker et al., 2011; Desai and Jin, 2011; Golubov et al., 2020). However, recent evidence that investor capital is sticky (Brunnermeier and Nagel, 2008; Koijen and Yogo, 2019) raises the question of why such catering arises. If shareholders rarely reconsider their holdings, it is not obvious why firms would adjust payout policy to appeal to them. This paper answers that question by identifying the condition under which the benefit of catering is more likely to be present. I find that firms cater when shareholders are reconsidering their portfolios. This finding suggests that catering decisions are determined not only by how strong investors' dividend preferences are, but also by when investor capital is

mobile.

Second, this paper provides novel evidence that fund flows have real effects through the portfolio reallocation they induce. Prior work emphasizes flow-induced price pressure (Edmans et al., 2012; Khan et al., 2012; Lou and Wang, 2018) or stock price fragility (Friberg et al., 2024; Massa et al., 2025). Neither channel accounts for the symmetric dividend increases I document. Price pressure implies opposite responses, because inflows and outflows push prices in opposite directions. Fragility predicts lower dividends, since firms exposed to fragile ownership hold more cash as a buffer against future price shocks. In contrast, the pattern is consistent with firms responding to reallocation that is common to both directions: large inflows and outflows push fund managers into discretionary trading. Identifying this reallocation channel broadens our understanding of how fund flows affect corporate policy.

Third, this paper contributes to the literature on flow-induced trading by showing that fund managers' response to flows varies with flow magnitude. Previous studies treat flow-induced trading as largely proportional across existing positions (Coval and Stafford, 2007; Lou, 2012), though recent evidence suggests that trading under extreme flows is informed and selective (Berger, 2023; Huang et al., 2023). I find that discretionary trading increases with the size of flows, suggesting that proportional allocation describes fund managers' response to typical flows, while the deliberate decisions documented in recent work emerge as flows become extreme.

Fourth, this paper contributes to the literature on investor attention and corporate behavior. In the existing literature, attention has been understood as disciplinary pressure on managers. Changes in monitoring intensity caused by limited attention lead to dividend changes (Kempf et al., 2017; Iliev et al., 2021). In contrast, I consider a different role for attention, which creates an incentive for managers to appeal to shareholders. The results are consistent with firms satisfying shareholders' preferences for dividends rather than firms mitigating agency frictions.

In sum, the full set of evidence highlights a strategic interaction in which large fund flows induce active portfolio rebalancing, firms respond by increasing dividends, and funds subsequently reallocate capital toward firms that raise dividends. Underlying this interaction is likely a shift in investor attention: rather than scaling their positions mechanically and proportionally, fund managers facing large flows actively reconsider what they hold. The paper thus identifies a distinct setting in which investors pay closer attention to their portfolio firms. Most prior research has examined circumstances in which attention is dispersed, so that investors fail to monitor their holdings (Cohen and Frazzini, 2008; Hirshleifer, Lim, et al., 2009; Schmidt, 2019). More recent evidence, however, suggests that certain events concentrate attention instead, prompting investors to process firm-specific news (Hirshleifer and Sheng, 2022). Large fund flows appear to play a similar role. If such periods indeed concentrate investor attention, they may motivate

other investor-oriented actions, such as voluntary disclosure choices or investor relations activities. In other words, the timing logic documented here can extend beyond payout policy.

The remainder of this paper is structured as follows. Section 2 describes the data and sample. Section 3 explains how I measure discretionary trading intensity and relates it to the size of mutual fund flows. Section 4 presents the dividend response to flow-induced portfolio reallocation. Section 5 provides causal evidence from the introduction of the Morningstar Sustainability Rating. Section 6 shows the cross-sectional evidence on catering. Section 7 presents fund responses to strategic dividend increases. Section 8 concludes.

## 2 Data and Sample Construction

### 2.1 Data Sources

My empirical analysis relies on three primary data sources. First, I obtain firm-level accounting information and stock market data for a global sample of publicly traded companies from Worldscope and Datastream. From Worldscope, I retrieve key balance sheet, income statement, and cash flow items, including total assets, debt, capital expenditures, R&D expenditures, cash holdings, intangible assets, and operating cash flows, along with corporate payout data comprising dividends and share repurchases. From Datastream, I collect firm-level stock market data, including stock prices, returns, shares outstanding, and dividend announcement dates.

Second, I source mutual fund characteristics and performance from Morningstar Direct, focusing on actively managed open-ended equity funds domiciled in various countries. In addition to total net assets (TNA), returns, inception dates, and fund domiciles, I retrieve each fund’s Morningstar category and its Morningstar Sustainability Rating, first published in March 2016, which I use in the difference-in-differences analysis in Section 5.

Third, I retrieve institutional equity positions from the FactSet Ownership database. This database provides a comprehensive repository of equity positions held by various institutions, defined as professional money managers with discretionary authority over assets, including mutual funds, pension funds, and insurance companies. All monetary variables, including firm-level accounting items, stock prices, fund-level TNAs, and fund flows, are converted into U.S. dollar terms.

## 2.2 Sample Construction

### 2.2.1 Mutual Fund Sample

To construct the fund sample, I merge the fund-level characteristics from Morningstar Direct with the position-level holdings from FactSet, which yields a comprehensive dataset of portfolio allocations for my sample funds. I limit the universe to open-ended equity funds, identified using Morningstar’s Broad Category Group classification. For funds with multiple share classes, I aggregate TNAs across all classes to obtain the fund-level TNA, and calculate the fund-level return as the average of the individual share class returns. Following Elton et al. (2001) and Evans (2010), I restrict the sample to funds that have been in existence for at least three years and manage TNA of at least \$15 million to mitigate incubation and survivorship biases. Because FactSet reports institutional equity positions on a quarterly basis, the fund-level panel is analyzed at a quarterly frequency.<sup>3</sup>

The baseline sample consists of actively managed funds, whose managers make the discretionary reallocation decisions that motivate my analysis. I exclude passive funds, defined as funds that Morningstar classifies as index funds or whose fund names contain the words “index,” “idx,” or “passive.” Although excluded from the baseline sample, these passive funds are used in a placebo test, reported in Table A4 in the Appendix and discussed in Section 4.2. After these screens, the baseline sample contains 14,593 funds across 43 domicile countries, corresponding to 500,195 fund-quarter observations spanning 2000:Q1 through 2021:Q4.

### 2.2.2 Firm Sample

To construct the baseline sample for the firm-level analysis, I apply a series of standard screening procedures. Following Massa et al. (2021), I employ the filters recommended by Ince and Porter (2006) to ensure data integrity. Specifically, I retain primary listings of common equity with non-missing country information and remove penny stocks, extreme return reversals, and stale prices. Next, I exclude firms in the utilities (SIC 4900–4999) and financial (SIC 6000–6999) sectors because of their distinct regulatory environments and capital structures. Firms are included in the empirical analysis only if they have non-missing values for the main payout variables and all control variables. Because flow-induced trading pressure is defined only for firms that are covered by FactSet and held by at least one sample fund, I further require firms to satisfy both conditions. These screens

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<sup>3</sup>To address the heterogeneous reporting frequencies (quarterly, semi-annual, or annual) in FactSet, I standardize the data to a quarterly frequency using a forward-fill approach, assuming that missing quarters carry over the latest reported holdings. To maintain data integrity, I do not apply forward-filling if the reporting gap exceeds four consecutive quarters, or if the fund reaches its final reporting date in the database. This prevents distorting the data by misinterpreting fund liquidations or data dropouts as continued holdings.



yield a baseline sample of 17,853 firms from 68 countries, corresponding to 147,110 firm-year observations covering the years from 2000 to 2021.

## 2.3 Summary Statistics

Table 1 reports summary statistics for the fund and firm samples. All continuous variables are winsorized at the 1st and 99th percentiles. Panel A covers fund-quarter observations. The average mutual fund in the sample manages \$0.8 billion in TNA, earns a quarterly return of 2.7%, and is approximately 15 years old. Average net flows are slightly negative over the sample period, at  $-0.4\%$  of beginning-of-quarter TNA. Although these average net flows are close to zero, the magnitude of flows remains economically large. The average absolute net flow is approximately 7.6% of beginning-of-quarter TNA, suggesting that mutual funds continue to experience substantial capital movements even when inflows and outflows largely offset each other.

Panel B covers the firm-level sample used in the baseline panel regressions. Firms distribute, on average, 2.8% of beginning-of-period assets through dividends and share repurchases. Dividends account for the majority of total payouts, at approximately 2% of assets, whereas repurchases are more concentrated among fewer firms: the median firm pays a positive dividend but repurchases no shares. The sample firms exhibit characteristics consistent with growth firms, with a mean market-to-book ratio of 2.6 and a median of 1.6, broadly in line with prior corporate finance studies using international samples of listed firms (e.g., Pinkowitz et al., 2016). On average, firms invest approximately 8% of beginning-of-period assets and have debt and cash ratios of 22% and 21%, respectively. Intangible assets represent 13% of total assets, and the average return on equity is 4.9%. Institutional ownership averages 17% across firms.

## 3 Flow-Induced Portfolio Reassessment

A central premise of this paper is that large inflows or outflows lead fund managers to reassess their portfolio strategies and reconsider how capital is allocated across portfolio firms. Moderate flows do not necessarily have this effect: prior work shows that managers accommodate a substantial fraction of flows by scaling existing positions proportionally (Lou, 2012). Large flows are different. As liquidity and price impact constraints bind, managers need to decide which positions to adjust and by how much. This portfolio reassessment is critical to the paper’s argument because it creates a window of opportunity during which firms may influence mutual funds’ subsequent investment decisions. Before examining whether firms exploit this window through payout policies, I test this premise directly by investigating whether larger fund flows are associated with greater discretionary trading activity.



### 3.1 Discretionary Trading Intensity

Portfolio reassessment is not directly observable. Thus, I infer it from the extent to which realized fund trading deviates from a benchmark of purely proportional trading. If managers simply accommodate investor flows by scaling existing positions, trading should closely follow a pro-rata allocation based on existing portfolio weights. In contrast, if managers revisit their portfolio strategies, they are more likely to selectively increase some positions while reducing others, generating trading patterns that deviate from proportional allocation. To capture such behavior, I construct a measure of Discretionary Trading Intensity (*DTI*).

I follow three steps to construct this measure. First, I calculate the fund's realized trading, defined as the quarterly change in the number of shares of each individual stock held by the fund:

$$Trading_{f,s,q} = \Delta Holding_{f,s,q}$$

where  $Holding_{f,s,q}$  is the number of shares of stock  $s$  held by fund  $f$  at the end of quarter  $q$ . Utilizing the number of shares allows me to eliminate confounding effects caused by stock price fluctuations and isolate the trading activities themselves.

Second, I compute pro-rata trading, which represents the hypothetical trading that would occur under the assumption that the fund manager allocates fund flows in proportion to existing portfolio holdings:

$$ProRataTrading_{f,s,q} = Holding_{f,s,q-1} \times Flow_{f,q}$$

where flows for fund  $f$  in quarter  $q$  are defined as

$$Flow_{f,q} = \frac{TNA_{f,q} - (1 + Ret_{f,q})TNA_{f,q-1}}{TNA_{f,q-1}}$$

and  $TNA_{f,q}$  is total net assets for fund  $f$  as of the end of quarter  $q$ , and  $Ret_{f,q}$  is the quarterly return for fund  $f$  in quarter  $q$ .

Third, I assemble these two trading components into a fund-level measure of discretionary trading intensity. For each fund, I aggregate the absolute value of the difference between realized trading and pro-rata trading across all stocks held by the fund. Since the difference is measured in the number of shares, I multiply it by the stock price at the end of the previous quarter to convert it into a dollar value.

$$DTI_{f,q} = \sum_s (|Trading_{f,s,q} - ProRataTrading_{f,s,q}| \times Price_{s,q-1}) \quad (1)$$

I use absolute values because the focus is on the magnitude of discretionary trading rather

than its direction.<sup>4</sup> Accordingly, any buying or selling that deviates from the predicted pro-rata trading is interpreted as reflecting the fund manager’s discretionary portfolio decision.

This approach offers two distinct advantages. First, by design, the measure disentangles the passive component of trading from its active counterpart and thus captures the degree to which fund managers actively expand or reduce stock positions beyond flow-induced proportional trading. Second, because the calculation does not divide by the stock price, it avoids building in a mechanical correlation between trading and returns (e.g., Wardlaw, 2020).

Panel A of Table 1 reports summary statistics for  $DTI$ . Discretionary trading is economically meaningful. In the median fund-quarter, the fund executes \$41.9 million of trading beyond what would be predicted under purely proportional allocation of investor flows. This amount corresponds to approximately a quarter of the median fund’s TNA. The measure varies widely across fund-quarters, ranging from \$11.2 million at the 25th percentile to \$155.3 million at the 75th percentile. I exploit this variation in the tests that follow.

### 3.2 Fund Flows and Discretionary Trading

Figure 1 provides a first look at how discretionary trading varies with the size of investor flows. Fund-quarters are sorted into terciles of net flows, and the figure plots the average ratio of  $DTI$  to TNA within each tercile. The relation is U-shaped. Discretionary trading is lowest in the middle tercile, where flows are close to zero, at 28.3% of TNA, and rises to 33.6% in the bottom tercile and 38.4% in the top tercile. The increase on both sides of the distribution suggests that discretionary trading depends on the magnitude of flows rather than their direction, which motivates using absolute flows as the explanatory variable in the following analysis.

To examine whether larger fund flows are associated with greater discretionary trading by mutual fund managers, I estimate the following regression model:

$$\ln(DTI)_{f,q} = \beta_1 |Flow|_{f,q} + \beta_2 \ln(TNA)_{f,q-1} + \beta_3 Ret_{f,q-1} + \alpha_f + \alpha_{cat,q} + \varepsilon_{f,q} \quad (2)$$

where  $DTI_{f,q}$  denotes the discretionary trading intensity of fund  $f$  in quarter  $q$ . The main explanatory variable,  $|Flow|_{f,q}$ , measures the magnitude of investor inflows and outflows.

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<sup>4</sup>The summation runs over the union of stocks held by fund  $f$  at the end of quarter  $q - 1$  and quarter  $q$ , so that newly initiated and fully liquidated positions are included. For a stock the fund did not hold at  $q - 1$ , the pro-rata benchmark is zero and the entire purchase is counted as discretionary trading. For a position the fund exits during quarter  $q$ , realized trading is  $-Holding_{f,s,q-1}$  while the benchmark is  $Holding_{f,s,q-1} \times Flow_{f,q}$ , so the deviation is the sale in excess of what proportional scaling would predict. This treatment is intentional: decisions to open or close a position involve a substantial degree of managerial discretion.

I use contemporaneous flows to capture fund managers' trading adjustments at the time investor flows are realized. The control variables include lagged fund size and lagged fund returns. The specification also includes fund fixed effects and category-by-year-quarter fixed effects, so that  $\beta_1$  is estimated from within-fund variation in flows relative to other funds in the same investment category and quarter. This addresses the concern that the univariate pattern in Figure 1 reflects differences across funds rather than variation in flows within a given fund. Standard errors are two-way clustered at the fund and category-by-year-quarter levels.

The coefficient of interest is  $\beta_1$ . If larger fund flows induce fund managers to reassess their portfolio strategies and engage in more active trading beyond proportional allocation,  $\beta_1$  is expected to be positive. In contrast, if fund flows primarily generate mechanical scaling of existing positions,  $\beta_1$  should be statistically insignificant.

Table 2 presents the relation between absolute fund flows and discretionary trading intensity. Across all specifications, the coefficient on  $|Flow|$  is positive and statistically significant at the 1% level, indicating that funds experiencing larger inflows or outflows engage in more discretionary trading activity. The economic magnitude is also substantial. Using the full-sample estimate in Column (1), a one-standard-deviation increase in absolute fund flows is associated with an approximately 32% increase in discretionary trading intensity. Larger flows are thus accompanied by greater deviations from proportional flow allocations, which is what one would expect if they lead fund managers to reconsider their portfolios.

Columns (2) and (3) split the sample by the direction of flows. The positive association between absolute fund flows and discretionary trading intensity is present in both the inflow and outflow subsamples, indicating that funds with large flows deviate from pro-rata trading patterns regardless of the direction of investor flows. The estimate is larger for inflows than for outflows (2.67 versus 1.83), broadly consistent with Lou (2012), who shows that outflow-related trading is relatively more mechanical. Because  $DTI$  is measured in dollars, one concern is that it mechanically increases with fund size. Although the specification already controls for lagged fund size, Columns (4) and (5) further address this concern by scaling  $DTI$  by beginning-of-period TNA before taking logs. The estimates, 2.60 for inflows and 1.82 for outflows, are nearly identical to their unscaled counterparts, so the relation is not simply driven by larger funds trading larger dollar amounts.<sup>5</sup>

The estimated relation between lagged returns and discretionary trading provides

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<sup>5</sup>Table A2 in the Appendix confirms that these results do not depend on how discretionary trading is normalized. Panel A scales  $DTI$  by beginning-of-period TNA, and Panel B scales each holding's discretionary trading by the firm's shares outstanding before aggregating to the fund level. Panel C uses an indicator for whether the ratio of  $DTI$  to the fund's total trading falls in the top quintile of the sample, which addresses the concern that the results simply reflect funds with higher total trading volume. The coefficient on  $|Flow|$  is positive and statistically significant in every specification. Table A3 further shows that larger flows are associated with actual changes in portfolio weights.

additional insight into when fund managers are more likely to deviate from proportional allocation. Lagged returns are negatively associated with discretionary trading intensity, and the association is concentrated in the outflow subsample. This result is consistent with fund managers reconsidering their portfolios more actively after weak performance, suggesting that discretionary trading reflects deliberate portfolio decisions.

Overall, the evidence supports the central premise of the paper. As the magnitude of investor flows increases, fund managers deviate more from proportional allocation of those flows, and they do so following both inflows and outflows. These findings suggest that large flow realizations identify periods in which portfolio firms are more likely to be reevaluated by their mutual fund shareholders. These periods provide the window of opportunity that motivates the analysis of corporate payout responses in the next section.

## 4 Dividend Responses to Portfolio Reallocation

Section 3 suggests that large flows create periods in which portfolio weights are actively revisited. Sections 4 and 5 examine whether firms increase dividends during these periods, first in the full panel and then using a quasi-experiment. This section presents the panel evidence.

Firms have reason to care about these periods. They take actions to make their shares attractive to investors (Covrig et al., 2007; Bushee and Miller, 2012). The payoff to those actions depends on whether investors respond by moving capital, which they rarely do (Brunnermeier and Nagel, 2008; Kojen and Yogo, 2019). When large flows force fund managers to reassess their portfolios, corporate policy is more likely to affect their investment decisions. Therefore, a firm that recognizes these periods has an incentive to act within them.

Dividend policy is a natural place to observe such a response for two reasons. First, the dividend catering literature has documented that firms adjust dividends to cater to shareholder preferences (Baker and Wurgler, 2004; Becker et al., 2011; Golubov et al., 2020). Second, dividends are sticky and costly to reverse (Lintner, 1956; Brav et al., 2005), so timing a dividend increase to a particular window is likely to reflect a deliberate decision. I hypothesize that firms strategically increase dividends when their mutual fund shareholders experience large flows.

### 4.1 Measuring Exposure to Flow-Induced Reallocation

To test the hypothesis developed above, I construct a firm-level measure of exposure to flow-induced portfolio reallocation. Because Section 3.2 shows that larger flows are associated with greater discretionary trading, firms whose mutual fund owners experience larger flows are more likely to have their positions reconsidered. This motivates the

following measure of flow-induced trading pressure:

$$Trading\ Pressure_{s,q} = \frac{\sum_f Holding_{f,s,q-1} \times |Flow|_{f,q}}{Shares\ Outstanding_{s,q-1}} \quad (3)$$

where  $Holding_{f,s,q-1}$  denotes the number of shares of stock  $s$  held by fund  $f$  at the end of quarter  $q - 1$ ,  $|Flow|_{f,q}$  is the absolute value of fund  $f$ 's flow during quarter  $q$ , and  $Shares\ Outstanding_{s,q-1}$  is the number of shares outstanding for stock  $s$  at the end of quarter  $q - 1$ . The measure is constructed quarterly and annualized by aggregating over the firm's fiscal year for the annual panel regressions.

A firm receives a high value of *Trading Pressure* when a large fraction of its shares is held by funds experiencing substantial inflows or outflows. The use of absolute values is motivated by the evidence in Section 3.2, where discretionary trading rises with the magnitude of flows regardless of their direction. The measure therefore captures the extent to which the firm's shareholder base is exposed to flow-induced portfolio reallocation.

Large flows may also affect firms through a separate channel: temporary price pressure. The existing literature shows that mutual fund inflows and outflows generate demand shocks that push prices away from fundamentals (e.g., Coval and Stafford, 2007; Edmans et al., 2012). To distinguish this mechanism from portfolio reallocation, I construct a complementary measure that retains the sign of flows:

$$Price\ Pressure_{s,q} = \frac{\sum_f Holding_{f,s,q-1} \times Flow_{f,q}}{Shares\ Outstanding_{s,q-1}} \quad (4)$$

The two measures differ only in how they treat the sign of flows. *Trading Pressure* captures the intensity of flow-induced reallocation regardless of direction, whereas *Price Pressure* captures the net demand shock imposed on the stock. They yield different predictions. If firms respond to reallocation, the response should be similar for inflows and outflows, since both prompt managers to revisit their weights. If firms respond to prices, the response should have opposite signs for inflows and outflows, since inflows push prices up and outflows push them down. Comparing the two measures helps distinguish between these explanations.

Panel B of Table 1 reports summary statistics. *Trading Pressure* averages 1.66% of lagged shares outstanding. In a typical firm-year, the absolute flows experienced by the firm's mutual fund shareholders, weighted by their holdings, represent 1.66% of the firm's shares. Exposure varies considerably in the cross section. The median firm has a value of 0.82%, while the firm at the 75th percentile has 2.54%, against a standard deviation of 2.00%. *Price Pressure*, by contrast, averages  $-0.15\%$ , close to zero and slightly negative, mirroring the mildly negative average net flow in Panel A. Signed flows largely cancel across a firm's shareholders, whereas their magnitudes accumulate.

Using the pressure measures, I estimate the following panel regression:

$$Y_{s,t} = \beta \textit{Trading Pressure}_{s,t-1} + \Gamma' \textit{Controls}_{s,t-1} + \alpha_s + \alpha_{c,i,t} + \varepsilon_{s,t} \quad (5)$$

where  $Y_{s,t}$  is dividends or share repurchases for firm  $s$  in fiscal year  $t$ , each scaled by beginning-of-period total assets, and  $\alpha_s$  and  $\alpha_{c,i,t}$  denote firm and country  $\times$  industry  $\times$  year fixed effects, respectively. Some specifications include *Price Pressure* $_{s,t-1}$ , constructed over the same window. Both pressure measures are standardized to have a mean of zero and a standard deviation of one, so coefficients indicate the effect of a one-standard-deviation increase in exposure. The hypothesis is  $\beta > 0$  for dividends, and the null hypothesis is  $\beta = 0$ .

Controls include standard determinants of payout policy, namely firm size, leverage, market-to-book, profitability, institutional ownership, cash holdings, intangible assets, investment, and selling, general, and administrative expenses. Variable definitions appear in Table A1 in the Appendix. Since the specification includes firm fixed effects and country  $\times$  industry  $\times$  year fixed effects,  $\beta$  is identified from within-firm variation in exposure after controlling for shocks common to firms in the same country, industry, and year. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels.

## 4.2 Baseline Results: Dividends and Reallocation

Table 3 reports the baseline estimates. Column (1) shows that the coefficient on *Trading Pressure* is positive and statistically significant at the 5% level. A one-standard-deviation increase in exposure corresponds to an increase in dividends of 0.03 percentage points of assets, or approximately 1.5% of the unconditional mean dividend. Column (2) adds *Price Pressure*. The coefficient on *Price Pressure* is essentially zero and statistically insignificant, and the coefficient on *Trading Pressure* is unchanged. This finding suggests that dividends respond to the extent of shareholder reallocation, rather than to the net demand pressure that those flows impose on the stock.

Column (3) decomposes *Trading Pressure* into inflow- and outflow-induced components, constructed from funds experiencing positive and negative flows, respectively. Both coefficients are positive and statistically significant, and their magnitudes are nearly identical. Firms increase dividends when their shareholders are reallocating, whether the underlying funds experience large inflows or large outflows. This symmetry is difficult to reconcile with a price-based explanation, since inflows and outflows exert opposite pressure on stock prices. Instead, firms appear to respond to the reallocation itself.

Column (4) turns to repurchases, which behave differently. The coefficient on the outflow-induced trading pressure is positive and statistically significant at the 1% level, whereas the coefficient on the inflow-induced component is smaller and statistically in-

significant. Because the outflow component is, by construction, the absolute value of the negative side of *Price Pressure*, this asymmetry is consistent with the predictions of a price-based explanation: outflows depress prices and improve the terms on which a firm can buy back its own shares, while inflows do the opposite. Repurchases thus reflect the price consequences of flows, consistent with a market timing motive, while dividends are driven by the reallocation.<sup>6</sup>

If firms respond to reallocation, the dividend response should be strongest among firms exposed to funds with extreme flows, because reallocation rises with the magnitude of flows as shown in Section 3.2. Table 4 tests this prediction directly by splitting *Trading Pressure* into a component attributable to funds with extreme flows, defined as quarterly flows above the 90th or below the 10th percentile of the flow distribution in that quarter, and a component attributable to funds with normal-range flows. Column (1) includes only the extreme component, Column (2) only the normal-range component, and Column (3) both. The coefficient on the extreme component is positive and statistically significant at the 1% level in both columns in which it appears, while the coefficient on the normal-range component is indistinguishable from zero. The dividend response is concentrated where flows are large enough to disrupt existing allocations.

Column (4) decomposes each component by the direction of flows. The coefficients on both the extreme inflow- and extreme outflow-induced pressures are positive and statistically significant, but the coefficients on the normal-range pressures are not. The symmetry documented in Table 3 holds within extreme flows and is absent outside them, which is consistent with the prediction of the reallocation interpretation.

A further test distinguishes active from passive funds. Passive fund managers exercise little discretion over portfolio decisions, so their flows involve little of the active portfolio reallocation that motivates the analysis. If firms respond to reallocation rather than to other effects of flows, the symmetry documented above should be a feature of active flows alone. Table A4 in the Appendix confirms this prediction. Trading pressure from active funds raises dividends following both inflows and outflows. By contrast, pressure from passive funds produces an asymmetric response. The coefficient on the outflow component is positive and statistically significant, while the coefficient on the inflow component is indistinguishable from zero. The symmetric response thus appears among active funds, whose flows involve reallocation, and disappears among passive funds, whose flows largely do not.<sup>7</sup>

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<sup>6</sup>Untabulated results confirm this interpretation directly. Regressing repurchases on *Price Pressure* in place of *Trading Pressure* yields a negative and statistically significant coefficient, so firms repurchase more when their shareholders' flows impose net selling pressure on the stock.

<sup>7</sup>The estimates from passive funds look like the repurchase pattern in Column (4) of Table 3, which is what a price-based explanation predicts. This is consistent with passive outflows generating mechanical selling pressure. Why dividends respond to such selling pressure is an interesting open question. Signaling is one possibility, but I leave this question for future research.



### 4.3 The Timing of Dividend Announcements

If firms deliberately adjust dividends around periods of reallocation, the response should show up in the timing of announcements and not only in dividend levels. Figure 2 provides a first look. Panel A restates the premise of Section 3 using quintiles of absolute fund flows: discretionary trading rises monotonically from 27.3% of TNA in the bottom quintile to 46.1% in the top quintile. Panel B turns to the firms these funds hold. Among firms that raise their dividend in a given fiscal year, I sort firm-quarters into quintiles of flow-induced trading pressure in the previous quarter. The share of dividend announcements rises monotonically across quintiles, from 15.2% in the bottom quintile to 26.8% in the top quintile.

Table 5 tests the timing prediction formally at a quarterly frequency. The dependent variable, an indicator for whether the firm announces a dividend in quarter  $q$ , is regressed on *Trading Pressure* in quarters  $q$  through  $q - 3$ . The specification includes firm  $\times$  year and country  $\times$  industry  $\times$  year-quarter fixed effects, so the estimates are identified from variation in announcement timing across quarters within the same firm-year.

Column (1) shows that exposure in the immediately preceding quarter predicts dividend announcements: a one-standard-deviation increase in  $Trading\ Pressure_{q-1}$  raises the probability of an announcement by 0.23 percentage points. This effect is statistically significant at the 1% level. The coefficient on exposure in the contemporaneous quarter is positive but smaller and statistically insignificant, and the coefficients on exposure two and three quarters earlier are negative and statistically insignificant. The one-quarter lag is what a deliberate response implies.<sup>8</sup> A mechanical or spurious correlation between flows and payout would not produce a response concentrated at exactly one lag and absent at the others.

Columns (2) through (4) examine which types of dividend announcements drive the result. Each firm-year is classified by whether the firm's dividend for the fiscal year is higher than, equal to, or lower than its dividend in the prior year, and each column restricts the sample to firm-quarters in firm-years of the corresponding type. The effect is driven by announcements of dividend increases, where a one-standard-deviation increase in lagged exposure raises the announcement probability by 0.35 percentage points. Announcements that leave dividends unchanged and decrease dividends show no statistically significant relation. Rather than simply announcing more often, firms exposed to large flows announce dividend increases.

Overall, the results in this section suggest that firms increase dividends in response to the reallocation that large flows induce. Three patterns support this interpretation.

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<sup>8</sup>The insignificant contemporaneous coefficient is not surprising, since  $Trading\ Pressure_q$  aggregates flows over the entire quarter and so may include flows realized after the announcement date. Dividend decisions also require board approval and time to prepare, so the response to a flow shock is more likely to appear in the following quarter than in the same one.

First, firms held by funds experiencing large flows pay higher dividends. Second, the response is symmetric across inflows and outflows. Third, the response is concentrated in dividend increases announced in the quarter immediately following the flows.

## 5 Evidence from the Morningstar Globe Rating

The evidence in Section 4 is consistent with firms timing dividend increases to flow-induced portfolio reallocation, but it does not establish a causal relation. Fund flows are not randomly assigned. They are associated with past fund performance and may therefore be correlated with the characteristics of the firms a fund holds, so that flow exposure and dividend policy could be jointly determined by an omitted factor. This section addresses that concern by exploiting a shock that moves fund flows for reasons unrelated to the fundamentals of the funds' portfolio firms.

On March 1, 2016, Morningstar introduced its sustainability rating system, assigning a discrete number of “Globes,” ranging from 1 to 5, to more than 20,000 mutual funds representing over \$8 trillion in aggregate assets under management. The ratings were constructed from pre-existing portfolio holdings and historical firm-level sustainability scores from Sustainalytics. As a result, the public release of these visual ratings altered investor perceptions and capital allocation without any contemporaneous change in the fundamentals of either the funds or the firms they hold. This feature makes the introduction a plausibly exogenous shock to fund flows. The flow responses were economically meaningful: funds receiving the highest rating (Five Globes) experienced substantial net inflows, whereas funds receiving the lowest rating (One Globe) experienced substantial net outflows (Hartzmark and Sussman, 2019). This two-sided response makes the setting well suited to my question. The rating introduction generated inflow and outflow shocks within a single event, allowing me to test the symmetry of the dividend response while holding the source of variation fixed.

Implementing this strategy requires fund-level sustainability ratings around the 2016 introduction. Morningstar currently provides ratings only from 2018 onward, because the methodology was revised in 2018 and earlier ratings are no longer publicly available.<sup>9</sup> Thus, I assign each fund its earliest available rating. This approach assumes that funds did not substantially change their underlying sustainability strategies in response to the methodological revision, an assumption supported by Gantchev et al. (2024), who find little evidence that funds strategically adjust their behavior when the rating methodology

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<sup>9</sup>The 2018 revision introduced several methodological changes, including the incorporation of historical portfolio sustainability scores, the expansion of peer comparison groups, and the introduction of a 1% buffer zone for rating upgrades and downgrades. Under this buffer rule, a fund's score must move at least 1% beyond the rating threshold for a rating change to occur. These changes are intended to reduce incentives for short-term manipulation of sustainability ratings.

changes.<sup>10</sup>

I construct the treatment and control groups from firms' ownership structure immediately prior to the shock, measured as of 2015:Q4. The treatment group consists of firms in the top quartile of ownership by either five-globe or one-globe funds. Firms with greater pre-shock ownership by these funds are more exposed to the flow shocks generated by the rating introduction. Pooling the two groups is motivated by the evidence in Figure 1 and Table 2, which show that active portfolio reallocation is closely related to the magnitude of flows rather than to their direction. I also estimate the effect separately for firms exposed to five-globe funds and firms exposed to one-globe funds to test whether the dividend response is symmetric across inflow- and outflow-driven reallocation. The control group consists of firms without substantial ex-ante ownership by either five-globe or one-globe funds.

Using these groups, I estimate the following difference-in-differences (DiD) specification:

$$Dividend_{s,t} = \beta Treat_s \times Post_t + \Gamma'(\bar{X}_{s,pre} \times \tau_t) + \alpha_s + \alpha_{c,i,t} + \varepsilon_{s,t} \quad (6)$$

where  $Dividend_{s,t}$  denotes dividends for firm  $s$  in fiscal year  $t$ , scaled by beginning-of-period total assets.  $Treat_s$  is an indicator equal to one for firms in the treatment group and zero otherwise, and  $Post_t$  is an indicator equal to one for fiscal years after 2015 and zero otherwise.  $\bar{X}_{s,pre}$  is a vector of the firm characteristics used in Table 3, averaged over the 2013 to 2015 period, and  $\tau_t$  is a vector of year indicators. Interacting these pre-event characteristics with year indicators controls for their differential effects on dividends over the event window. The specification includes firm fixed effects and country  $\times$  industry  $\times$  year fixed effects. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels.

The event window spans the two fiscal years before the introduction (2014 and 2015) and the three fiscal years after it (2016 through 2018). Fiscal year 2016 is assigned to the post period because the ratings were released in March 2016, early enough for dividend decisions during that fiscal year to respond. Table 5 supports this assignment. Firms adjust dividend announcements in the quarter immediately following a flow shock, so the response to the shock realized in the first quarter of 2016 would occur within fiscal year 2016. The coefficient of interest is  $\beta$ , which captures the differential change in dividends for treated firms following the rating introduction. Given the panel evidence, the predicted sign of  $\beta$  is positive.

Table 6 reports the DiD results. In Column (1), the coefficient on  $Treat \times Post$  is

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<sup>10</sup>Table A5 in the Appendix confirms that this assignment recovers the shock. Columns (1) and (2) show that, relative to other funds, five-globe funds receive significantly larger inflows after 2016:Q1, while one-globe funds experience significant outflows, reproducing the pattern in Hartzmark and Sussman (2019). Columns (3) and (4) further show that both groups are about 2.3 percentage points more likely to fall in the top quintile of the ratio of discretionary trading to total trading after the introduction, which is consistent with the flow shock prompting the portfolio reassessment documented in Section 3.2.

positive and statistically significant at the 1% level. Treated firms raise dividends by 0.24 percentage points of assets relative to control firms, an economically meaningful adjustment for a payout policy that firms are reluctant to revise (Lintner, 1956; Brav et al., 2005). Column (2) replaces *Post* with year-specific indicators, where *Pre*<sub>2015</sub> denotes the year before the introduction and *Post*<sub>2016</sub>, *Post*<sub>2017</sub>, and *Post*<sub>2018</sub> denote the year of the introduction and the two years that follow. The coefficient on *Treat*  $\times$  *Pre*<sub>2015</sub> is small and statistically insignificant, providing no evidence of differential pre-trends and supporting the parallel trends assumption. The three post-treatment coefficients are all positive, statistically significant at the 1% level, and similar in size, moving from 0.0023 in 2016 to 0.0028 in 2017 and 2018. As anticipated in the design of the event window, the response is already present in the year of the introduction, and it persists thereafter.<sup>11</sup>

Column (3) estimates the effect separately for the inflow and outflow shocks by splitting the treatment group into firms whose ownership by five-globe funds is in the top quartile and firms whose ownership by one-globe funds is in the top quartile. The former captures exposure to funds receiving inflows and the latter to funds experiencing outflows. Both coefficients are positive and statistically significant, at 0.0016 for the five-globe group and 0.0020 for the one-globe group. Dividends increase whether the shock to a firm’s shareholder base is an inflow or an outflow, echoing the symmetry in Column (3) of Table 3. What the two groups share is not the direction of the shock, which moves prices in opposite ways, but their shareholders’ reassessment of portfolios. Firms raise dividends when that reassessment is under way. Furthermore, Table A6 in the Appendix traces the dynamics separately for the two groups and shows that neither displays a differential pre-trend.

Taken together, the evidence in this section supports a causal interpretation of the panel results: firms respond to flow-driven portfolio reassessments by increasing their dividends. Firms whose shareholders are most exposed to the rating-induced flow shocks raise dividends relative to control firms, and the effect appears in the year of the shock and persists thereafter. The response is symmetric across the inflow and outflow shocks.

## 6 Catering to Reallocating Shareholders

Sections 4 and 5 show that firms increase dividends when their shareholders are reconsidering their portfolios. This section argues that the dividend response reflects catering, in which firms raise dividends to satisfy their shareholders’ preferences for dividends (Baker and Wurgler, 2004; Li and Lie, 2006; Golubov et al., 2020). A dividend increase is more

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<sup>11</sup>Table A7 in the Appendix shows that the increase is reflected in dividend decisions to raise or initiate a dividend. Among firms that were paying dividends as of 2015, treated firms are 5.1 percentage points more likely to raise their dividend in the year of the introduction, and among firms that were not paying dividends, treated firms are 5.2 percentage points more likely to initiate a dividend in 2017. Consistent with Column (2), the effects appear only in the post-introduction years.

likely to shape shareholders’ allocations when they are revisiting their portfolios, making these periods a natural time to cater. If firms are catering, the response should be more pronounced among those with a stronger catering incentive.

To test the prediction, I employ the difference-in-differences design of Section 5. Each specification interacts  $Treat \times Post$  with an indicator for firms with a stronger incentive to cater. The indicator equals one for firms above the sample median of the corresponding characteristic described in Section 6.1. All controls and fixed effects are further interacted with the group indicator, so the treatment effect is estimated separately within each group and the interaction term reports the difference between them. Because every group indicator is constructed from characteristics measured as of 2015, group assignment is determined before the rating introduction and cannot respond to the shock.

## 6.1 Evidence of Catering

A firm’s incentive to cater is stronger when its shareholders place a higher value on dividends. I measure the strength of this preference with three proxies based on the composition of a firm’s mutual fund ownership: ownership by income funds, ownership by funds domiciled in countries that cut dividend taxes in 2015, and ownership by funds domiciled in countries with older populations. In each case, ownership is measured as of 2015 as the holdings of the relevant fund group divided by the firm’s total mutual fund holdings.

The first proxy infers dividend preferences directly from funds’ investment objectives. Income funds are defined as equity funds whose names contain “dividend,” “income,” or “yield.” These are the shareholders whose allocation decisions heavily weight payout. For a firm whose shareholder base is tilted toward them, a dividend increase during a reallocation window is more likely to be rewarded with capital. Motivated by this idea, I sort firms on the share of their mutual fund ownership held by income funds. Column (1) of Table 7 reports the result. The interaction with  $D(High Own_{Income Funds})$  is 0.0033 and significant at the 5% level. Firms with a strong dividend clientele raise dividends by an additional 0.33 percentage points of assets, roughly three times the effect for other firms.

The second proxy exploits cross-country variation in dividend taxation. The value investors place on dividends depends on how dividends are taxed (Allen et al., 2000; Desai and Jin, 2011), and firms adjust payouts when dividend taxes change (Chetty and Saez, 2005). A cut in dividend taxes raises the after-tax value of dividends to investors in the affected country and should therefore strengthen the dividend preference of the funds that serve them. I identify countries that reduced their overall dividend tax rates in 2015<sup>12</sup> and sort firms on the share of their mutual fund ownership held by funds domiciled

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<sup>12</sup>Dividend tax rates are from the OECD Tax Database, which reports the overall statutory tax rate

in these countries. Column (2) shows that the interaction with  $D(High\ Own_{Tax\ Cut})$  is 0.0025 and significant at the 5% level. Firms with high ownership by these funds raise dividends by an additional 0.25 percentage points of assets, about one and a half times the effect for other firms. The tax cut itself may raise dividends among firms held by these funds regardless of the rating introduction. However, because the country  $\times$  industry  $\times$  year fixed effects are interacted with the group indicator, any common change in dividends within the group is absorbed. The interaction therefore captures only the additional response of treated firms within the group.

The third proxy exploits the demographic composition of the funds' domiciles. Older investors exhibit a stronger preference for dividends (Graham and Kumar, 2006), and firms located in areas with a larger share of older residents are more likely to pay dividends (Becker et al., 2011). Following the United Nations, I classify a country as having an older population if at least 14% of its residents are aged 65 or over. I then sort firms on the share of their mutual fund ownership held by funds domiciled in such countries. Column (3) shows a similar result to Columns (1) and (2). The interaction with  $D(High\ Own_{Older\ Populations})$  is 0.0022 and significant at the 10% level. This corresponds to an additional 0.22 percentage points of assets, nearly twice the effect for other firms.

The three proxies are complementary. Ownership by income funds measures dividend preferences most directly, because these funds explicitly state an objective of generating income for their investors. However, income funds choose which firms to hold, so they may select firms that are likely to increase dividends. The second and third proxies address this concern. They capture dividend preferences less directly, through the tax treatment and demographics of fund investors, but a country's tax policy and age structure are plausibly unrelated to the characteristics of any individual portfolio firm.

Across all three tests, the dividend response during reallocation windows is significantly stronger among firms whose shareholders strongly prefer dividends. The evidence indicates that the dividend increases documented in Sections 4 and 5 reflect a catering motive.

## 6.2 Alternative Explanation: Governance

A remaining concern is that dividends rise during reallocation windows for governance reasons. In the agency view of payout, distributing cash limits corporate managers' discretion over free cash flow and subjects them to the scrutiny of external capital markets (Jensen, 1986; Easterbrook, 1984). Large flows plausibly strengthen these forces. When

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on dividend income, combining corporate and personal taxes. A country is classified as having cut its dividend tax if this rate declined in 2015. Because tax rates are available only for countries covered by the database, ownership for this proxy is measured relative to the holdings of funds domiciled in these countries. The database covers 34 fund domiciles in the sample, of which seven reduced their rates in 2015.



fund managers are actively reviewing which positions to adjust, portfolio firms come under closer monitoring and the threat of exit becomes more likely to be exercised (Edmans, 2009; McCahery et al., 2016). A firm facing this pressure may raise its dividend as a commitment not to waste its cash. The testable implication is that the response should be concentrated among firms where free cash flow problems are most severe.

I use the same design as in Table 7, replacing the indicators with proxies for the severity of free cash flow problems. I use two sets of proxies. The first captures how much cash is available for managers to spend. Highly levered firms are already disciplined by debt service, so low leverage indicates a more severe agency problem. Large cash balances and high free cash flow, measured as operating cash flow net of investment, indicate greater discretionary resources. The second set captures investment opportunities, since retaining cash is harder to justify when profitable uses for it are scarce. The proxies for this set are the market-to-book ratio, sales growth, and realized investment, with low values indicating limited opportunities. If governance drives the dividend response, the interaction should be positive and significant for each of the six indicators.

Table 8 shows that none of the six interaction terms is statistically significant. The point estimates are also small, and all three proxies for investment opportunities show the opposite sign: the interactions with  $D(Low\ M/B)$ ,  $D(Low\ Sales\ Growth)$ , and  $D(Low\ Investment)$  are negative, indicating, if anything, a weaker response among firms with limited investment opportunities. Furthermore, the lack of significance is not a matter of statistical power. The interaction standard errors, between 0.0012 and 0.0013, are essentially the same as those in Table 7, where the same design detects significant heterogeneity. Thus, governance does not appear to explain the dividend response.

## 7 Fund Responses to Dividend Increases

The evidence so far shows that firms time dividend increases to reallocation windows and that they do so to cater to their shareholders' preferences for dividends. This section examines whether the strategy works. If catering during these periods works as intended, funds that are reconsidering their portfolios should tilt toward firms that raise dividends.

The test uses a fund-firm-quarter panel built from the funds whose flows are moved by the rating introduction, namely those receiving five globes and those receiving one globe. These are the funds that are reconsidering their portfolios, so they are where a response to dividend increases should appear. I estimate

$$\begin{aligned} Trading_{f,s,q} = & \beta_1 Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0) \\ & + \beta_2 Treat_s \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0) \\ & + \Gamma'(\bar{X}_{s,pre} \times \tau_q) + \alpha_s + \alpha_{f,q} + \alpha_{Treat \times Post} + \varepsilon_{f,s,q} \end{aligned}$$



where  $Trading_{f,s,q}$  is fund  $f$ 's quarterly trading in stock  $s$ , scaled by the firm's lagged shares outstanding.  $Treat_s$  identifies the treated firms of Table 6, those most exposed to the rating-induced flow shocks.  $I(\Delta Dividend_{2015-2016} > 0)$  equals one for firms that raised their dividend between 2015 and 2016, the increases documented in Section 5.  $Post_{>2016}$  equals one for quarters after the end of firm  $s$ 's 2016 fiscal year. The indicator is firm-specific because fiscal year ends differ across firms. Defining it this way ensures that the post period begins after the firm's 2016 dividend is known to investors. The specification includes firm fixed effects and fund  $\times$  year-quarter fixed effects. I also include  $Treat \times Post$  fixed effects to control for any differential change in trading toward treated firms after the shock that is unrelated to their dividend decisions. Following Table 6, pre-event average firm characteristics interacted with year-quarter dummies are also included. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels.

The coefficient of interest is  $\beta_2$ . It asks whether the payoff to raising dividends (i.e., capital from affected funds) is larger for firms more heavily exposed to shareholders' portfolio reallocation. Because the fund  $\times$  year-quarter fixed effects absorb the fund's aggregate response to its own flows,  $\beta_2$  is identified from how a fund allocates across firms *within* its portfolio at a moment when it is deciding which positions to adjust.

Table 9 reports the results. In Column (1), which pools five-globe and one-globe funds, the triple interaction is 0.0030 and significant at the 1% level. Funds facing the rating-induced flow shock trade toward treated firms that raise dividends, buying an additional 0.3% of shares outstanding per quarter. This finding shows that the dividend increases documented in Section 5 do not merely coincide with reallocation windows, but also translate into capital allocation.

Columns (2) and (3) estimate the effect separately for five-globe and one-globe funds. The triple interaction is positive and significant in both, at 0.0056 for funds receiving inflows and 0.0025 for funds experiencing outflows. Since the two groups face opposite shocks, they make different decisions. A fund with new capital chooses where to put it, while a fund facing redemptions chooses what to sell and what to keep. Firms that raise dividends are favored in both cases. A dividend increase during a reallocation window thus both attracts capital from funds deploying new money and retains capital in funds meeting redemptions.

Taken with Sections 5 and 6, the evidence suggests a strategic interaction. Large flows force fund managers to revisit their portfolios. Firms whose mutual fund shareholders experience large flows raise dividends to cater to their shareholders' preferences for dividends. Funds then allocate toward the firms that raised dividends.

## 8 Conclusion

Firms adjust dividends to make their shares attractive to investors. Yet investors rarely move their capital, and they do not constantly reconsider their portfolios. This paper examines whether firms increase dividends when their shareholders are actively reallocating their portfolios. I first identify these periods using large mutual fund flows. Fund managers deviate further from proportional flow allocation as flows grow larger, and they do so following both inflows and outflows. Large flows create windows in which portfolio weights are actively revisited.

Firms increase dividends in these windows. Firms whose mutual fund shareholders receive large flows pay higher dividends, and the response is concentrated among firms exposed to extreme flows. The response is symmetric across inflows and outflows. This symmetry suggests that firms respond to the flow-induced reassessment of portfolios, not to the price impact of the flows. These dividend increases are announced in the quarter immediately following the flows, suggesting that firms strategically choose the timing. A difference-in-differences design exploiting the 2016 introduction of Morningstar Sustainability Ratings supports a causal interpretation.

The cross-sectional evidence indicates that firms cater to their shareholders' preferences for dividends. The response is stronger among firms with greater ownership by income funds, by funds domiciled in countries that cut dividend taxes, and by funds domiciled in countries with older populations. It is not stronger among firms with more severe free cash flow problems, so governance does not appear to explain the response.

Funds do respond to these increases. Following the rating release, both funds receiving inflows and funds facing outflows trade toward firms that raised dividends, so the dividend increases effectively attract new capital and retain existing capital.

Overall, the findings suggest that firms and their shareholders strategically interact around portfolio reallocation: firms time their payout to these moments, and shareholders allocate capital based on that payout. Dividend catering illustrates a broader pattern in which firms anticipate when investors will act and choose their actions accordingly. Uncovering other actions firms take, and other moments at which they take those actions, is a promising direction for future research.

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Figure 1: Net Fund Flows and Discretionary Trading

Figure 1 plots average discretionary trading intensity, scaled by total net assets, across terciles of net fund flows. Fund-quarters are sorted into terciles by net flows, with “Low” denoting the bottom third of the flow distribution and “High” the top third.

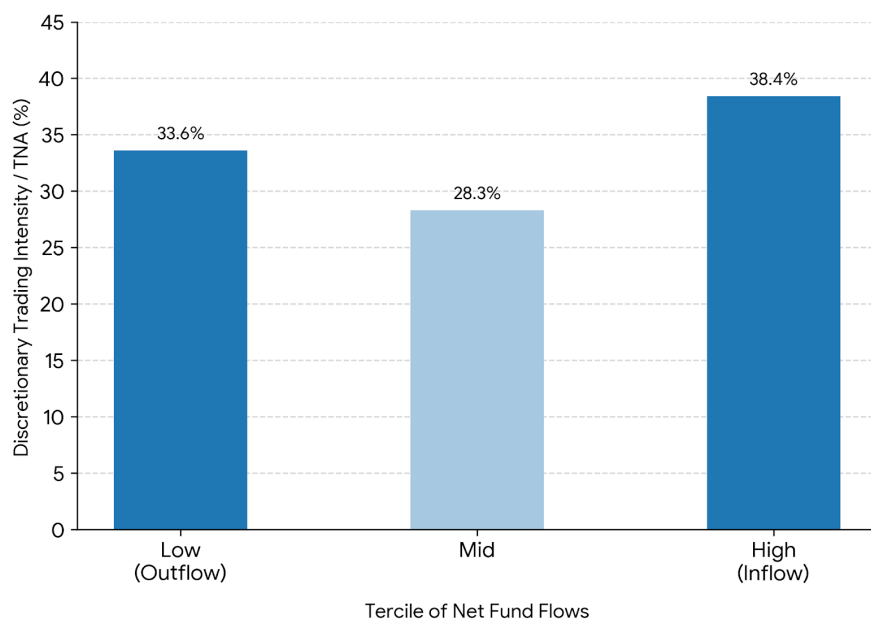


Figure 2: Fund Flows, Discretionary Trading, and Dividend-Increase Announcements

Panel A plots average discretionary trading intensity, scaled by beginning-of-quarter total net assets, across quintiles of absolute fund flows. The sample is the fund-quarter sample used in Column (1) of Table 2. Panel B restricts the firm-quarter sample of Table 5 to firm-years in which the firm raises its dividend, sorts firm-quarters into quintiles of flow-induced trading pressure in the previous quarter, and plots the share of all dividend announcements in these firm-years that falls in each quintile. The dashed line marks 20%, the share each quintile would receive if announcements were evenly distributed across quintiles.

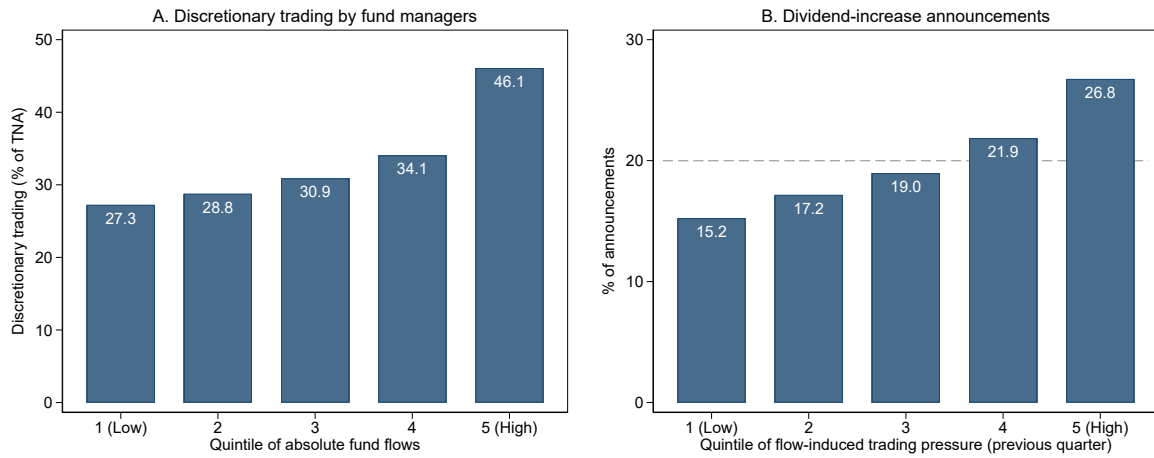




Table 1: Summary Statistics

Table 1 reports the summary statistics for the funds and firms used in the panel regressions. I report the mean, standard deviation, 25th percentile (P25), median, and 75th percentile (P75) for each variable. Panel A presents the summary statistics for funds. The fund sample consists of actively managed equity funds with total net assets of at least \$15 million and at least three years of history. Panel B presents the summary statistics for the firm sample, excluding all financial (SIC 6000–6999) and utilities (SIC 4900–4999) companies. Variable definitions are provided in Table A1 in the Appendix. The sample period spans from 2000 to 2021. All continuous variables are winsorized at the 1% level in both tails.

| Panel A: Fund Sample    |         |         |         |         |         |         |
|-------------------------|---------|---------|---------|---------|---------|---------|
|                         | Obs     | Mean    | STD     | P25     | Median  | P75     |
| <i>DTI</i> (\$bil)      | 500,195 | 0.2009  | 0.4788  | 0.0112  | 0.0419  | 0.1553  |
| $\ln(DTI)$              | 500,195 | -3.2365 | 1.9872  | -4.4959 | -3.1730 | -1.8622 |
| <i>Flow</i>             | 500,195 | -0.0042 | 0.1274  | -0.0516 | -0.0188 | 0.0179  |
| $ Flow $                | 500,195 | 0.0764  | 0.1114  | 0.0186  | 0.0406  | 0.0840  |
| <i>TNA</i> (\$bil)      | 500,195 | 0.7603  | 1.7422  | 0.0619  | 0.1824  | 0.6027  |
| <i>Return</i>           | 500,195 | 0.0267  | 0.1011  | -0.0193 | 0.0336  | 0.0835  |
| <i>Age</i> (years)      | 500,195 | 15.1691 | 10.3654 | 8       | 13      | 19.75   |
| Panel B: Firm Sample    |         |         |         |         |         |         |
|                         | Obs     | Mean    | STD     | P25     | Median  | P75     |
| <i>Dividend</i>         | 147,110 | 0.0196  | 0.0318  | 0       | 0.0087  | 0.0237  |
| <i>Repurchase</i>       | 147,110 | 0.0087  | 0.0263  | 0       | 0       | 0.0015  |
| <i>Trading Pressure</i> | 147,110 | 0.0166  | 0.0200  | 0.0016  | 0.0082  | 0.0254  |
| <i>Price Pressure</i>   | 147,110 | -0.0015 | 0.0113  | -0.0046 | -0.0002 | 0.0010  |
| <i>Assets</i> (\$bil)   | 147,110 | 3.2700  | 8.6603  | 0.2107  | 0.6090  | 2.0604  |
| <i>Debt</i>             | 147,110 | 0.2217  | 0.2021  | 0.0413  | 0.1859  | 0.3434  |
| <i>M/B</i>              | 147,110 | 2.5677  | 3.0200  | 0.9229  | 1.6038  | 2.9316  |
| <i>ROE</i>              | 147,110 | 0.0486  | 0.2784  | 0.0186  | 0.0836  | 0.1576  |
| <i>IO</i>               | 147,110 | 0.1670  | 0.1599  | 0.0373  | 0.1112  | 0.2575  |
| <i>Cash</i>             | 147,110 | 0.2094  | 0.2104  | 0.0646  | 0.1437  | 0.2798  |
| <i>Intangible</i>       | 147,110 | 0.1321  | 0.2023  | 0.0057  | 0.0338  | 0.1758  |
| <i>Investment</i>       | 147,110 | 0.0814  | 0.0890  | 0.0248  | 0.0542  | 0.1038  |
| <i>SGA/Sale</i>         | 147,110 | 0.3498  | 0.7917  | 0.1025  | 0.1897  | 0.3364  |

Table 2: Absolute Fund Flows and Discretionary Trading Intensity

Table 2 presents the results of regressions examining the relation between Discretionary Trading Intensity ( $DTI$ ) and the magnitude of fund flows.  $DTI$  is defined as the sum of the absolute deviations of actual trading from hypothetical trading in which fund flows are allocated proportionally to existing portfolio weights. The dependent variable is the natural logarithm of  $DTI$  in Columns (1)–(3) and the natural logarithm of  $DTI$  scaled by beginning-of-period total net assets ( $TNA$ ) in Columns (4)–(5). Section 3 describes the measure in more detail. The key independent variable,  $|Flow|_t$ , is the absolute value of net fund flows during the quarter. Control variables are lagged fund size ( $\ln(TNA)_{t-1}$ ) and lagged fund returns ( $Return_{t-1}$ ). Column (1) uses the full sample, Columns (2) and (4) restrict to fund-quarters with positive flows, and Columns (3) and (5) to those with negative flows. All specifications include fund and category  $\times$  year-quarter fixed effects. Standard errors are two-way clustered at the fund and category  $\times$  year-quarter levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                | $\ln(DTI)$             |                       |                        | $\ln(DTI/TNA)$         |                        |
|-----------------------------------|------------------------|-----------------------|------------------------|------------------------|------------------------|
|                                   | Full                   | $Flow > 0$            | $Flow < 0$             | $Flow > 0$             | $Flow < 0$             |
| Sample                            | (1)                    | (2)                   | (3)                    | (4)                    | (5)                    |
| $ Flow _t$                        | 2.4896***<br>(0.0319)  | 2.6741***<br>(0.0301) | 1.8270***<br>(0.0476)  | 2.5959***<br>(0.0273)  | 1.8172***<br>(0.0471)  |
| $\ln(TNA)_{t-1}$                  | 0.8859***<br>(0.0052)  | 0.8797***<br>(0.0067) | 0.9057***<br>(0.0060)  | -0.0974***<br>(0.0059) | -0.0794***<br>(0.0055) |
| $Return_{t-1}$                    | -0.1770***<br>(0.0663) | -0.1199<br>(0.0953)   | -0.4825***<br>(0.0788) | -0.0755<br>(0.0941)    | -0.4739***<br>(0.0787) |
| Observations                      | 500,195                | 168,931               | 329,748                | 168,931                | 329,748                |
| Adjusted R-squared                | 0.7278                 | 0.7246                | 0.7353                 | 0.3887                 | 0.3272                 |
| Fund FE                           | Y                      | Y                     | Y                      | Y                      | Y                      |
| Category $\times$ Year-Quarter FE | Y                      | Y                     | Y                      | Y                      | Y                      |

Table 3: Flow-Induced Trading Pressure and Payout Policy

Table 3 reports regression estimates of the relation between flow-induced trading pressure and corporate payout policy for the global sample of firms from Worldscope. The dependent variable is either *Dividend* (Columns (1)–(3)) or *Repurchase* (Column (4)). The key independent variables are measures of firm-level exposure to mutual fund flows. Specifically, *Trading Pressure* (*Price Pressure*) is calculated as the sum, across the firm’s mutual fund shareholders, of the product of each fund’s lagged holdings in the firm and the absolute (signed) value of its net flows, scaled by the firm’s lagged shares outstanding. Section 4.1 provides a detailed description of these measures. All pressure measures are standardized to have a mean of zero and a standard deviation of one. In Columns (3) and (4), *Trading Pressure* is decomposed into *Trading Pressure<sub>inflow</sub>* and *Trading Pressure<sub>outflow</sub>* based on the sign of fund flows. All specifications include a set of firm-level controls, as well as firm and country  $\times$  industry  $\times$  year fixed effects. Industry classifications follow the Fama-French 48 industries. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                         | <i>Dividend</i><br>(1) | <i>Dividend</i><br>(2) | <i>Dividend</i><br>(3) | <i>Repurchase</i><br>(4) |
|--------------------------------------------|------------------------|------------------------|------------------------|--------------------------|
| <i>Trading Pressure</i>                    | 0.0003**<br>(0.0001)   | 0.0003**<br>(0.0001)   |                        |                          |
| <i>Price Pressure</i>                      |                        | 0.0000<br>(0.0001)     |                        |                          |
| <i>Trading Pressure<sub>inflow</sub></i>   |                        |                        | 0.0002**<br>(0.0001)   | 0.0001<br>(0.0001)       |
| <i>Trading Pressure<sub>outflow</sub></i>  |                        |                        | 0.0002*<br>(0.0001)    | 0.0006***<br>(0.0002)    |
| $\ln(Assets)$                              | -0.0002<br>(0.0003)    | -0.0002<br>(0.0003)    | -0.0002<br>(0.0003)    | 0.0037***<br>(0.0004)    |
| <i>Debt</i>                                | -0.0197***<br>(0.0009) | -0.0197***<br>(0.0009) | -0.0197***<br>(0.0009) | -0.0164***<br>(0.0010)   |
| <i>M/B</i>                                 | 0.0011***<br>(0.0001)  | 0.0011***<br>(0.0001)  | 0.0011***<br>(0.0001)  | 0.0004***<br>(0.0001)    |
| <i>ROE</i>                                 | 0.0102***<br>(0.0005)  | 0.0102***<br>(0.0005)  | 0.0102***<br>(0.0005)  | 0.0059***<br>(0.0005)    |
| <i>IO</i>                                  | 0.0076***<br>(0.0015)  | 0.0076***<br>(0.0015)  | 0.0073***<br>(0.0015)  | -0.0045***<br>(0.0017)   |
| <i>Cash</i>                                | 0.0090***<br>(0.0008)  | 0.0090***<br>(0.0008)  | 0.0090***<br>(0.0008)  | 0.0087***<br>(0.0009)    |
| <i>Intangible</i>                          | -0.0052***<br>(0.0010) | -0.0052***<br>(0.0010) | -0.0052***<br>(0.0010) | -0.0105***<br>(0.0011)   |
| <i>Investment</i>                          | -0.0004<br>(0.0016)    | -0.0004<br>(0.0016)    | -0.0004<br>(0.0016)    | -0.0054***<br>(0.0015)   |
| <i>SGA/Sale</i>                            | 0.0001<br>(0.0002)     | 0.0001<br>(0.0002)     | 0.0001<br>(0.0002)     | -0.0002<br>(0.0002)      |
| Observations                               | 147,110                | 147,110                | 147,110                | 147,110                  |
| Adjusted R-squared                         | 0.7259                 | 0.7259                 | 0.7259                 | 0.3893                   |
| Firm FE                                    | Y                      | Y                      | Y                      | Y                        |
| Country $\times$ Industry $\times$ Year FE | Y                      | Y                      | Y                      | Y                        |

Table 4: Dividend Responses to Extreme Flow-Induced Trading Pressure

Table 4 examines whether corporate dividend responses to flow-induced trading pressure are concentrated among funds experiencing extreme flows. Columns (1)–(3) decompose *Trading Pressure* into components attributable to funds with extreme flows (*Trading Pressure<sup>extreme</sup>*) and funds with normal-range flows (*Trading Pressure<sup>normal</sup>*). Extreme flows are defined as quarterly fund flows above the 90th or below the 10th percentile of the flow distribution in a given quarter; all remaining flows are classified as normal. Column (4) further decomposes each component by the direction of fund flows. All trading pressure measures are standardized to have a mean of zero and a standard deviation of one. The dependent variable in all columns is dividends, scaled by beginning-of-period total assets. All specifications include a set of firm-level controls, as well as firm and country  $\times$  industry  $\times$  year fixed effects. Industry classifications follow the Fama-French 48 industries. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                                          | <i>Dividend</i>       |                     |                       |                       |
|-------------------------------------------------------------|-----------------------|---------------------|-----------------------|-----------------------|
|                                                             | (1)                   | (2)                 | (3)                   | (4)                   |
| <i>Trading Pressure<sup>extreme</sup></i>                   | 0.0003***<br>(0.0001) |                     | 0.0003***<br>(0.0001) |                       |
| <i>Trading Pressure<sup>normal</sup></i>                    |                       | -0.0000<br>(0.0002) | -0.0001<br>(0.0002)   |                       |
| <i>Trading Pressure<sup>extreme</sup><sub>inflow</sub></i>  |                       |                     |                       | 0.0002***<br>(0.0001) |
| <i>Trading Pressure<sup>extreme</sup><sub>outflow</sub></i> |                       |                     |                       | 0.0003***<br>(0.0001) |
| <i>Trading Pressure<sup>normal</sup><sub>inflow</sub></i>   |                       |                     |                       | -0.0001<br>(0.0001)   |
| <i>Trading Pressure<sup>normal</sup><sub>outflow</sub></i>  |                       |                     |                       | -0.0001<br>(0.0001)   |
| Observations                                                | 147,110               | 147,110             | 147,110               | 147,110               |
| Adjusted R-squared                                          | 0.7259                | 0.7259              | 0.7259                | 0.7260                |
| Controls                                                    | Y                     | Y                   | Y                     | Y                     |
| Firm FE                                                     | Y                     | Y                   | Y                     | Y                     |
| Country $\times$ Industry $\times$ Year FE                  | Y                     | Y                   | Y                     | Y                     |

Table 5: Trading Pressure and Dividend Announcement Timing

Table 5 examines whether firms time their dividend announcements in response to large flows. The dependent variable is an indicator for whether a firm announces a dividend during quarter  $q$ , regressed on *Trading Pressure* in quarters  $q$  through  $q - 3$ . Column (1) uses the full sample, while Columns (2)–(4) report estimates separately for dividend increases, no dividend changes, and dividend decreases, respectively. *Trading Pressure* is the sum, across the firm’s mutual fund shareholders, of each fund’s lagged holdings in the firm multiplied by the absolute value of its net flows, scaled by the firm’s lagged shares outstanding, and is standardized to have a mean of zero and a standard deviation of one. All specifications include firm  $\times$  year and country  $\times$  industry  $\times$  year-quarter fixed effects. Industry classifications follow the Fama-French 48 industries. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                                  | $1(\text{Dividend Announcement})_q$ |                          |                     |                          |
|-----------------------------------------------------|-------------------------------------|--------------------------|---------------------|--------------------------|
| Sample                                              | Full<br>(1)                         | Dividend Increase<br>(2) | No Change<br>(3)    | Dividend Decrease<br>(4) |
| <i>Trading Pressure</i> <sub><math>q</math></sub>   | 0.0010<br>(0.0008)                  | 0.0024<br>(0.0016)       | -0.0001<br>(0.0002) | -0.0006<br>(0.0027)      |
| <i>Trading Pressure</i> <sub><math>q-1</math></sub> | 0.0023***<br>(0.0008)               | 0.0035**<br>(0.0016)     | 0.0001<br>(0.0002)  | 0.0024<br>(0.0025)       |
| <i>Trading Pressure</i> <sub><math>q-2</math></sub> | -0.0010<br>(0.0008)                 | 0.0003<br>(0.0017)       | -0.0002<br>(0.0002) | -0.0037<br>(0.0026)      |
| <i>Trading Pressure</i> <sub><math>q-3</math></sub> | -0.0003<br>(0.0009)                 | -0.0010<br>(0.0017)      | -0.0001<br>(0.0002) | -0.0022<br>(0.0026)      |
| Observations                                        | 716,184                             | 309,848                  | 189,302             | 174,180                  |
| Adjusted R-squared                                  | 0.5302                              | 0.5504                   | 0.5706              | 0.4643                   |
| Firm $\times$ Year FE                               | Y                                   | Y                        | Y                   | Y                        |
| Country $\times$ Industry $\times$ Year-Quarter FE  | Y                                   | Y                        | Y                   | Y                        |

Table 6: Difference-in-Differences Estimates of Changes in Dividends

Table 6 presents the results of the difference-in-differences (DiD) analysis examining changes in corporate dividend policy following the introduction of the Morningstar Sustainability Rating. The regression model is specified as follows:

$$Y_{s,t} = \beta \text{ Treat}_s \times \text{Post}_t + \Gamma'(\bar{X}_{s,pre} \times \tau_t) + \alpha_s + \alpha_{c,i,t} + \varepsilon_{s,t}$$

where  $Y_{s,t}$  denotes dividends for firm  $s$  in fiscal year  $t$ . The treatment indicator,  $\text{Treat}_s$ , equals one for firms in the top quartile of ownership by either “five-globe” or “one-globe” funds as of December 2015, and zero otherwise. The  $\text{Post}_t$  variable is an indicator equal to one for the post-event period (2016–2018) and zero for the two years preceding the introduction (2014–2015). To account for potential confounding factors, the model includes a set of pre-event average firm characteristics ( $\bar{X}_{s,pre}$  measured over 2013–2015) interacted with year dummies ( $\tau_t$ ). All specifications include firm and country  $\times$  industry  $\times$  year fixed effects. In Column (2), I replace  $\text{Post}_t$  with a set of year-specific indicators –  $\text{Pre}_{2015}$ ,  $\text{Post}_{2016}$ ,  $\text{Post}_{2017}$ , and  $\text{Post}_{2018}$  – to trace the dynamic effects of the rating introduction. In Column (3), the treatment group is split into two:  $\text{Treat}_{FiveGlobes}$  includes firms in the top quartile of ownership by “five-globe” funds, and  $\text{Treat}_{OneGlobe}$  includes those in the top quartile of ownership by “one-globe” funds. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                             | <i>Dividend</i>       |                       |                       |
|------------------------------------------------|-----------------------|-----------------------|-----------------------|
|                                                | (1)                   | (2)                   | (3)                   |
| $\text{Treat} \times \text{Post}$              | 0.0024***<br>(0.0006) |                       |                       |
| $\text{Treat} \times \text{Pre}_{2015}$        |                       | 0.0005<br>(0.0007)    |                       |
| $\text{Treat} \times \text{Post}_{2016}$       |                       | 0.0023***<br>(0.0009) |                       |
| $\text{Treat} \times \text{Post}_{2017}$       |                       | 0.0028***<br>(0.0009) |                       |
| $\text{Treat} \times \text{Post}_{2018}$       |                       | 0.0028***<br>(0.0009) |                       |
| $\text{Treat}_{FiveGlobes} \times \text{Post}$ |                       |                       | 0.0016**<br>(0.0007)  |
| $\text{Treat}_{OneGlobe} \times \text{Post}$   |                       |                       | 0.0020***<br>(0.0006) |
| Observations                                   | 38,133                | 38,133                | 38,133                |
| Adjusted R-squared                             | 0.8173                | 0.8172                | 0.8173                |
| Pre-event controls $\times$ Year               | Y                     | Y                     | Y                     |
| Firm FE                                        | Y                     | Y                     | Y                     |
| Country $\times$ Industry $\times$ Year FE     | Y                     | Y                     | Y                     |

Table 7: Heterogeneity in Dividend Responses: Catering Channel

Table 7 investigates whether the dividend response to the treatment is stronger among firms with high catering incentives. Each specification interacts the treatment effect,  $Treat \times Post$ , with an indicator for the group of firms with high catering incentives. In Column (1),  $D(High Own_{Income Funds})$  equals one for firms with high ownership by income funds, defined as equity funds whose names contain “dividend,” “income,” or “yield.” In Column (2),  $D(High Own_{Tax Cut})$  equals one for firms with high ownership by funds domiciled in countries that reduced their dividend tax rates in 2015. In Column (3),  $D(High Own_{Older Populations})$  equals one for firms with high ownership by funds domiciled in countries with older populations, defined following the United Nations as countries where at least 14% of the population is aged 65 or older. In each column, ownership is measured as the holdings of the specified fund group divided by the combined holdings of that group and all other funds. Each indicator equals one for firms whose corresponding characteristic, measured as of 2015, is above the sample median, and zero otherwise. All specifications include the pre-event average firm characteristics in Table 6 interacted with year dummies, firm fixed effects, and country  $\times$  industry  $\times$  year fixed effects, each further interacted with the group indicator. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                         | Dividend             |                      |                     |
|--------------------------------------------|----------------------|----------------------|---------------------|
|                                            | (1)                  | (2)                  | (3)                 |
| $Treat \times Post$                        | 0.0012*<br>(0.0007)  | 0.0016**<br>(0.0008) | 0.0012*<br>(0.0007) |
| $\times D(High Own_{Income Funds})$        | 0.0033**<br>(0.0013) |                      |                     |
| $\times D(High Own_{Tax Cut})$             |                      | 0.0025**<br>(0.0012) |                     |
| $\times D(High Own_{Older Populations})$   |                      |                      | 0.0022*<br>(0.0011) |
| Observations                               | 35,595               | 35,212               | 35,236              |
| Adjusted R-squared                         | 0.8199               | 0.8212               | 0.8220              |
| Pre-event controls $\times$ Year           | Y                    | Y                    | Y                   |
| Firm FE                                    | Y                    | Y                    | Y                   |
| Country $\times$ Industry $\times$ Year FE | Y                    | Y                    | Y                   |

Table 8: Heterogeneity in Dividend Responses: Governance Channel

Table 8 examines whether the dividend response to the treatment is stronger among firms with greater free cash flow and weaker investment opportunities, where free cash flow problems are expected to be more severe. Each specification interacts  $Treat \times Post$  with an indicator for firms with low leverage (*Low Debt*), high cash holdings (*High Cash*), high free cash flow (*High FCF*), low market-to-book ratio (*Low M/B*), low sales growth (*Low Sales Growth*), or low investment (*Low Investment*). A high (low) indicator equals one for firms whose corresponding characteristic, measured as of 2015, is above (below) the sample median, and zero otherwise. The treatment indicator,  $Treat$ , equals one for firms in the top quartile of ownership by either five-globe or one-globe funds in 2015, and zero otherwise. The variable  $Post$  is an indicator equal to one for the post-event period (2016–2018) and zero for the pre-event period (2014–2015). All specifications include pre-event average firm characteristics in Table 6, interacted with year dummies, as well as firm and country  $\times$  industry  $\times$  year fixed effects. All controls and fixed effects are interacted with the group indicator. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                         | <i>Dividend</i>       |                      |                       |                       |                       |                       |
|--------------------------------------------|-----------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                            | (1)                   | (2)                  | (3)                   | (4)                   | (5)                   | (6)                   |
| $Treat \times Post$                        | 0.0031***<br>(0.0009) | 0.0023**<br>(0.0009) | 0.0022***<br>(0.0006) | 0.0029***<br>(0.0010) | 0.0026***<br>(0.0009) | 0.0031***<br>(0.0008) |
| $\times D(Low\ Debt)$                      | -0.0005<br>(0.0012)   |                      |                       |                       |                       |                       |
| $\times D(High\ Cash)$                     |                       | 0.0000<br>(0.0012)   |                       |                       |                       |                       |
| $\times D(High\ FCF)$                      |                       |                      | 0.0010<br>(0.0013)    |                       |                       |                       |
| $\times D(Low\ M/B)$                       |                       |                      |                       | -0.0013<br>(0.0013)   |                       |                       |
| $\times D(Low\ Sales\ Growth)$             |                       |                      |                       |                       | -0.0005<br>(0.0012)   |                       |
| $\times D(Low\ Investment)$                |                       |                      |                       |                       |                       | -0.0014<br>(0.0012)   |
| Observations                               | 32,630                | 32,438               | 32,405                | 32,560                | 32,592                | 32,559                |
| Adjusted R-squared                         | 0.8138                | 0.8150               | 0.8163                | 0.8151                | 0.8117                | 0.8152                |
| Pre-event controls $\times$ Year           | Y                     | Y                    | Y                     | Y                     | Y                     | Y                     |
| Firm FE                                    | Y                     | Y                    | Y                     | Y                     | Y                     | Y                     |
| Country $\times$ Industry $\times$ Year FE | Y                     | Y                    | Y                     | Y                     | Y                     | Y                     |



Table 9: Fund Trading Responses to Dividend Increases after the Rating Introduction

Table 9 reports regression estimates of the impact of dividend increases on funds' trading following the introduction of Morningstar Sustainability Ratings. The dependent variable, *Trading*, represents quarterly fund trading scaled by the firm's lagged shares outstanding. The treatment indicator, *Treat*, identifies the treated firms in Table 6.  $Post_{>2016}$  is an indicator variable equal to one for the quarters following the end of the firm's fiscal year 2016.  $I(\Delta Dividend_{2015-2016} > 0)$  is an indicator equal to one if a firm increases its dividend between 2015 and 2016 and zero otherwise. Column (1) pools five-globe and one-globe funds and uses a single treatment indicator, *Treat*. Columns (2) and (3) restrict the sample to five-globe funds and one-globe funds, respectively, and separately estimate the effects for  $Treat_{FiveGlobes}$  and  $Treat_{OneGlobe}$  groups. The triple interaction term,  $Treat \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$ , captures whether funds exhibit differential trading patterns toward treated firms that increase dividends following the release of ratings. All specifications include firm, fund  $\times$  year-quarter, and  $Treat \times Post$  fixed effects. Additionally, I control for pre-event average firm characteristics interacted with year-quarter dummies. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                                                                 | <i>Trading</i>         |                        |                        |
|------------------------------------------------------------------------------------|------------------------|------------------------|------------------------|
|                                                                                    | Five and One           | Five                   | One                    |
| Sample                                                                             | (1)                    | (2)                    | (3)                    |
| $Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$                           | -0.0032***<br>(0.0008) | -0.0055***<br>(0.0016) | -0.0023***<br>(0.0007) |
| $Treat \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$              | 0.0030***<br>(0.0010)  |                        |                        |
| $Treat_{FiveGlobes} \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$ |                        | 0.0056***<br>(0.0017)  |                        |
| $Treat_{OneGlobe} \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$   |                        |                        | 0.0025**<br>(0.0012)   |
| Observations                                                                       | 1,256,253              | 546,946                | 708,567                |
| Adjusted R-squared                                                                 | 0.0770                 | 0.0760                 | 0.0853                 |
| $Treat \times Post$                                                                | Y                      | Y                      | Y                      |
| Pre-event controls $\times$ Year-Quarter                                           | Y                      | Y                      | Y                      |
| Firm FE                                                                            | Y                      | Y                      | Y                      |
| Fund $\times$ Year-Quarter FE                                                      | Y                      | Y                      | Y                      |

## Appendix

This appendix contains additional analyses that support the main results of the paper. Table A1 provides definitions of the variables used in the paper. The remaining tables are organized around four sets of tests as follows:

### **Robustness of the flow-discretionary trading relation (Tables A2 and A3).**

Section 3.2 shows that discretionary trading increases with the magnitude of fund flows. Table A2 shows that this relation does not simply reflect fund size or overall trading volume. The coefficient on  $|Flow|$  remains positive and significant when  $DTI$  is scaled by total net assets or by shares outstanding, or replaced with an indicator for a high ratio of  $DTI$  to total trading. Table A3 addresses the concern that  $DTI$  may capture trades that merely restore existing portfolio weights. Larger flows are associated with lower overlap between a fund’s portfolio weights in consecutive quarters, following both inflows and outflows, indicating that funds actually change their portfolios.

**Placebo test using passive funds (Table A4).** Because passive funds track an index, their flows largely do not prompt active reallocation. If firms respond to reallocation itself, the symmetric dividend response should appear only for active funds. For passive funds, only the outflow component is significant, which resembles the repurchase results in Table 3 and is consistent with a response to underpricing.

**Validity of the natural experiment (Tables A5 and A6).** Table A5 confirms that assigning each fund its earliest available rating captures the 2016 rating shock. Five-globe funds receive significant inflows and one-globe funds significant outflows, as in Hartzmark and Sussman (2019), and both groups become more likely to have a high  $DTI$  ratio. Table A6 finds no differential pre-trend among firms exposed to either five-globe or one-globe funds.

**Dividend increases and initiations (Table A7).** Table A7 replaces the dependent variable in Table 6 with indicators for dividend increases and initiations. Treated firms that paid dividends in 2015 are about 5.1 percentage points more likely to raise them in 2016, and treated non-payers are about 5.2 percentage points more likely to initiate dividends in 2017, with no differential pre-trend in either case.

Table A1: Variable Definitions

| Variable                             | Definition                                                                                                                                                                                                                                                                   |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Panel A: Fund-level Variables</b> |                                                                                                                                                                                                                                                                              |
| <i>Flow</i>                          | The net flow to mutual fund $f$ during quarter $q$ is defined as: $Flow_{f,q} = \frac{TNA_{f,q} - (1 + Ret_{f,q})TNA_{f,q-1}}{TNA_{f,q-1}}$                                                                                                                                  |
| <i>DTI</i>                           | Discretionary trading intensity. The sum of absolute deviations of actual trading from a hypothetical trading benchmark, where investor flows are allocated proportionally based on existing portfolio weights.                                                              |
| <i>High DTI Ratio</i>                | An indicator equal to one if the ratio of <i>DTI</i> to the fund's total trading during the quarter is in the top 20 percent of the sample, and zero otherwise.                                                                                                              |
| <i>Overlap</i>                       | Portfolio similarity between two consecutive quarters. $Overlap_{f,q} = 1 - \frac{1}{2} \sum_i  w_{i,f,q} - w_{i,f,q-1} $ <p>where <math>w_{i,f,q}</math> is the weight of stock <math>i</math> in fund <math>f</math>'s portfolio at the end of quarter <math>q</math>.</p> |
| <b>Panel B: Firm-level Variables</b> |                                                                                                                                                                                                                                                                              |
| <i>Dividend</i>                      | Cash dividends scaled by the beginning-of-period total assets.                                                                                                                                                                                                               |
| <i>Repurchase</i>                    | Share repurchases scaled by the beginning-of-period total assets.                                                                                                                                                                                                            |
| <i>Trading Pressure</i>              | The sum of the product of each fund's lagged holdings in the firm and the absolute value of its net flows, scaled by the firm's lagged shares outstanding. $Trading\ Pressure_{s,q} = \frac{\sum_f Holding_{f,s,q-1} \times  Flow _{f,q}}{Shares\ Outstanding_{s,q-1}}$      |

Table A1 (continued)

| Variable              | Definition                                                                                                                                                                                                                                                               |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Price Pressure</i> | The sum of the product of each fund's lagged holdings in the firm and the signed value of its net flows, scaled by the firm's lagged shares outstanding.<br><br>$Price\ Pressure_{s,q} = \frac{\sum_f Holding_{f,s,q-1} \times Flow_{f,q}}{Shares\ Outstanding_{s,q-1}}$ |
| $\ln(Assets)$         | The natural logarithm of book value of total assets.                                                                                                                                                                                                                     |
| <i>Debt</i>           | The sum of long- and short-term debt scaled by the beginning-of-period total assets.                                                                                                                                                                                     |
| <i>M/B</i>            | Market-to-book ratio. The ratio of market value of equity to book value of equity.                                                                                                                                                                                       |
| <i>ROE</i>            | Return on equity.                                                                                                                                                                                                                                                        |
| <i>IO</i>             | Institutional ownership. The sum of all institutional holdings in FactSet divided by the number of shares outstanding.                                                                                                                                                   |
| <i>Cash</i>           | Total cash holdings divided by the beginning-of-period total assets.                                                                                                                                                                                                     |
| <i>Intangible</i>     | Intangible assets divided by the beginning-of-period total assets.                                                                                                                                                                                                       |
| <i>Investment</i>     | The sum of capital expenditures and R&D expenditures divided by the beginning-of-period total assets.                                                                                                                                                                    |
| <i>SGA/Sale</i>       | The ratio of selling, general, and administrative expenses to sales.                                                                                                                                                                                                     |
| <i>FCF</i>            | Free cash flow. The difference between operating cash flows and investment.                                                                                                                                                                                              |
| <i>Sales Growth</i>   | Growth rate of net sales.                                                                                                                                                                                                                                                |

Table A2: Alternative Measures of Discretionary Trading Intensity

Table A2 presents the results of regressions examining the relation between the magnitude of fund flows and three alternative measures of fund discretionary trading intensity. For dependent variables, Panel A uses  $DTI/TNA$ , which measures discretionary trading intensity relative to the fund's total net assets. Panel B uses  $DTI/SharesOutstanding$ , which is constructed by summing, at the fund level, the discretionary trading of each holding scaled by the firm's shares outstanding. Panel C uses an indicator for *High DTI Ratio*, which equals one if the ratio of  $DTI$  to the fund's total trading during the quarter is in the top 20 percent of the sample, and zero otherwise. The key independent variable,  $|Flow|$ , represents the absolute value of net fund flows during the quarter. Columns (1)–(3) report results for the full sample, funds experiencing positive flows, and funds experiencing negative flows, respectively. Control variables are lagged fund size ( $\ln(TNA)$ ) and lagged fund returns ( $Return$ ) as in Table 2. All specifications include fund and category  $\times$  year-quarter fixed effects. Standard errors are two-way clustered at the fund and category  $\times$  year-quarter levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Sample                                             | Full Sample<br>(1)    | $Flow > 0$<br>(2)     | $Flow < 0$<br>(3)     |
|----------------------------------------------------|-----------------------|-----------------------|-----------------------|
| <b>Panel A: <math>DTI/TNA</math></b>               |                       |                       |                       |
| $ Flow $                                           | 0.7839***<br>(0.0111) | 1.0337***<br>(0.0126) | 0.1434***<br>(0.0105) |
| Observations                                       | 500,201               | 168,931               | 329,754               |
| Adjusted R-squared                                 | 0.3888                | 0.4790                | 0.3530                |
| <b>Panel B: <math>DTI/SharesOutstanding</math></b> |                       |                       |                       |
| $ Flow $                                           | 1.3077***<br>(0.1347) | 1.6759***<br>(0.1716) | 0.4349**<br>(0.1744)  |
| Observations                                       | 500,186               | 168,924               | 329,746               |
| Adjusted R-squared                                 | 0.2152                | 0.2572                | 0.2089                |
| <b>Panel C: <i>High DTI Ratio</i></b>              |                       |                       |                       |
| $ Flow $                                           | 0.3622***<br>(0.0120) | 0.3937***<br>(0.0137) | 0.0605***<br>(0.0136) |
| Observations                                       | 414,726               | 138,800               | 274,081               |
| Adjusted R-squared                                 | 0.1913                | 0.1995                | 0.2248                |
| Controls                                           | Y                     | Y                     | Y                     |
| Fund FE                                            | Y                     | Y                     | Y                     |
| Category $\times$ Year-Quarter FE                  | Y                     | Y                     | Y                     |

Table A3: Absolute Fund Flows and Portfolio Similarity

Table A3 examines whether larger fund flows are associated with actual changes in portfolio weights. One concern with *DTI* is that it may capture trades that do not reflect portfolio reassessment. For example, when the price of a large holding rises, its weight in the portfolio increases, and a fund manager who rebalances to restore the original weights may deviate from the pro-rata benchmark. *DTI* would record such trades as discretionary even though the manager does not intend to change the portfolio. To address this concern, the dependent variable, *Overlap*, measures the similarity of a fund's portfolio weights between two consecutive quarters:  $Overlap_{f,q} = 1 - \frac{1}{2} \sum_i |w_{i,f,q} - w_{i,f,q-1}|$ , where  $w_{i,f,q}$  is the weight of stock  $i$  in fund  $f$ 's portfolio at the end of quarter  $q$ . *Overlap* equals one if the fund holds an identical portfolio in both quarters, and lower values indicate greater divergence. The key independent variable,  $|Flow|$ , is the absolute value of net fund flows during the quarter. Control variables are lagged fund size ( $\ln(TNA)$ ) and lagged fund returns (*Return*) as in Table 2. Column (1) uses the full sample, and Columns (2) and (3) restrict the sample to fund-quarters with positive and negative flows, respectively. All specifications include fund and category  $\times$  year-quarter fixed effects. Standard errors are two-way clustered at the fund and category  $\times$  year-quarter levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                | <i>Overlap</i>         |                        |                        |
|-----------------------------------|------------------------|------------------------|------------------------|
|                                   | Full                   | $Flow > 0$             | $Flow < 0$             |
| Sample                            | (1)                    | (2)                    | (3)                    |
| $ Flow _t$                        | -0.0728***<br>(0.0022) | -0.0578***<br>(0.0028) | -0.1041***<br>(0.0037) |
| $\ln(TNA)_{t-1}$                  | 0.0086***<br>(0.0005)  | 0.0097***<br>(0.0006)  | 0.0090***<br>(0.0006)  |
| $Return_{t-1}$                    | 0.0674***<br>(0.0057)  | 0.0271***<br>(0.0099)  | 0.0791***<br>(0.0073)  |
| Observations                      | 500,201                | 168,931                | 329,754                |
| Adjusted R-squared                | 0.2910                 | 0.3103                 | 0.2902                 |
| Fund FE                           | Y                      | Y                      | Y                      |
| Category $\times$ Year-Quarter FE | Y                      | Y                      | Y                      |

Table A4: Impact of Passive Fund Flows on Dividends

Table A4 compares firms' dividend responses to trading pressure from active and passive mutual fund shareholders. *Trading Pressure<sup>active</sup>* and *Trading Pressure<sup>passive</sup>* measure firms' exposure to the flows of their active and passive mutual fund shareholders, respectively, and are standardized. Passive funds are defined as funds classified as index funds by Morningstar or whose names contain the words "index," "idx," or "passive." Column (1) examines the effects of active and passive trading pressure separately. Column (2) further decomposes trading pressure by the direction of fund flows into inflow and outflow components. All specifications include a set of firm-level controls, as well as firm and country  $\times$  industry  $\times$  year fixed effects as in Table 3. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                                          | <i>Dividend</i>       |                       |
|-------------------------------------------------------------|-----------------------|-----------------------|
|                                                             | (1)                   | (2)                   |
| <i>Trading Pressure<sup>active</sup></i>                    | 0.0003**<br>(0.0001)  |                       |
| <i>Trading Pressure<sup>passive</sup></i>                   | 0.0004***<br>(0.0002) |                       |
| <i>Trading Pressure<sup>active</sup><sub>inflow</sub></i>   |                       | 0.0002**<br>(0.0001)  |
| <i>Trading Pressure<sup>active</sup><sub>outflow</sub></i>  |                       | 0.0002**<br>(0.0001)  |
| <i>Trading Pressure<sup>passive</sup><sub>inflow</sub></i>  |                       | 0.0000<br>(0.0001)    |
| <i>Trading Pressure<sup>passive</sup><sub>outflow</sub></i> |                       | 0.0005***<br>(0.0001) |
| Observations                                                | 152,135               | 152,135               |
| Adjusted R-squared                                          | 0.7246                | 0.7246                |
| Controls                                                    | Y                     | Y                     |
| Firm FE                                                     | Y                     | Y                     |
| Country $\times$ Industry $\times$ Year FE                  | Y                     | Y                     |

Table A5: Impact of Globe Rating on Fund Flows and Discretionary Trading

Table A5 reports the effects of the introduction of the Morningstar Sustainability Rating on mutual fund flows and discretionary trading. The dependent variable is net fund flow in Columns (1)–(2) and *High DTI Ratio* in Columns (3)–(4). *High DTI Ratio* equals one if the ratio of *DTI* to the fund’s total trading during the quarter is in the top 20 percent of the sample, and zero otherwise. The key independent variables are interaction terms between rating categories (*Five Globes* and *One Globe*) and a  $Post_{>2016Q1}$  indicator, which equals one for quarters after the first quarter of 2016 and zero otherwise. Rating categories are assigned based on the first available sustainability rating for each fund. The sample spans from 2015 to 2016. Columns (1) and (3) use the full sample, while Columns (2) and (4) restrict the sample to funds with sustainability ratings. All specifications include fund and category  $\times$  year-quarter fixed effects. Control variables include lagged fund size and lagged fund returns, each interacted with the  $Post_{>2016Q1}$  indicator. Standard errors are two-way clustered at the fund and category  $\times$  year-quarter levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                           | <i>Flow</i>           |                       | <i>High DTI Ratio</i> |                      |
|----------------------------------------------|-----------------------|-----------------------|-----------------------|----------------------|
|                                              | Full                  | Rated                 | Full                  | Rated                |
| Sample                                       | (1)                   | (2)                   | (3)                   | (4)                  |
| <i>Five Globes</i> $\times$ $Post_{>2016Q1}$ | 0.0111***<br>(0.0036) | 0.0088**<br>(0.0037)  | 0.0236**<br>(0.0111)  | 0.0235**<br>(0.0112) |
| <i>One Globe</i> $\times$ $Post_{>2016Q1}$   | -0.0073*<br>(0.0041)  | -0.0097**<br>(0.0042) | 0.0230**<br>(0.0111)  | 0.0228**<br>(0.0111) |
| Observations                                 | 65,573                | 58,845                | 57,881                | 52,323               |
| Adjusted R-squared                           | 0.2946                | 0.2920                | 0.2544                | 0.2565               |
| Controls                                     | Y                     | Y                     | Y                     | Y                    |
| Fund FE                                      | Y                     | Y                     | Y                     | Y                    |
| Category $\times$ Year-Quarter FE            | Y                     | Y                     | Y                     | Y                    |



Table A6: Dynamic Treatment Effects on Dividends by Inflow and Outflow Shocks

Table A6 reports the dynamic DiD estimates of dividend policy after the Morningstar Sustainability Rating introduction, separately for firms exposed to inflow or outflow shocks. The treatment variables are defined as follows:  $Treat_{FiveGlobes}$  includes firms in the top quartile of ownership by five-globe funds as of December 2015, and  $Treat_{OneGlobe}$  includes those in the top quartile of ownership by one-globe funds. Following Column (2) of Table 6, the  $Post_t$  indicator is decomposed into year-specific variables –  $Pre_{2015}$ ,  $Post_{2016}$ ,  $Post_{2017}$ , and  $Post_{2018}$  – to trace the dynamic effects of the rating introduction. All specifications include firm and country  $\times$  industry  $\times$  year fixed effects. Column (2) additionally includes pre-event average firm characteristics from 2013–2015, interacted with year dummies. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                         | <i>Dividend</i>       |                       |
|--------------------------------------------|-----------------------|-----------------------|
|                                            | (1)                   | (2)                   |
| $Treat_{FiveGlobes} \times Pre_{2015}$     | 0.0009<br>(0.0007)    | 0.0006<br>(0.0008)    |
| $Treat_{FiveGlobes} \times Post_{2016}$    | 0.0014<br>(0.0009)    | 0.0011<br>(0.0010)    |
| $Treat_{FiveGlobes} \times Post_{2017}$    | 0.0029***<br>(0.0009) | 0.0027**<br>(0.0011)  |
| $Treat_{FiveGlobes} \times Post_{2018}$    | 0.0018*<br>(0.0010)   | 0.0022*<br>(0.0012)   |
| $Treat_{OneGlobe} \times Pre_{2015}$       | -0.0002<br>(0.0006)   | -0.0001<br>(0.0006)   |
| $Treat_{OneGlobe} \times Post_{2016}$      | 0.0017**<br>(0.0007)  | 0.0020***<br>(0.0008) |
| $Treat_{OneGlobe} \times Post_{2017}$      | 0.0012<br>(0.0008)    | 0.0016**<br>(0.0008)  |
| $Treat_{OneGlobe} \times Post_{2018}$      | 0.0007<br>(0.0008)    | 0.0021**<br>(0.0008)  |
| Observations                               | 38,133                | 38,133                |
| Adjusted R-squared                         | 0.8162                | 0.8173                |
| Pre-event Controls $\times$ Year           | N                     | Y                     |
| Firm FE                                    | Y                     | Y                     |
| Country $\times$ Industry $\times$ Year FE | Y                     | Y                     |

Table A7: Dividend Increases and Initiations Following the Rating Introduction

Table A7 examines whether firms most exposed to the flow shock generated by the introduction of the Morningstar Sustainability Rating are more likely to raise or initiate dividend payments. The dependent variable, *Increase*, is an indicator equal to one if a firm raises its cash dividend relative to the prior fiscal year, and zero otherwise. *Initiation* is an indicator equal to one if a firm pays a dividend in the current fiscal year after paying none in the prior year, and zero otherwise. Columns (1) and (2) use the sample of firms paying dividends as of 2015, while Columns (3) and (4) use the sample of firms paying no dividends as of 2015. Columns (2) and (4) decompose *Post* into year-specific indicators to trace the dynamic effects of the rating introduction. All other specifications are identical to those in Table 6. Standard errors are two-way clustered at the firm and country  $\times$  industry  $\times$  year levels and reported in parentheses. \*\*\*, \*\*, and \* indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

| Dependent variable                         | <i>Increase</i>    | <i>Increase</i>     | <i>Initiation</i>  | <i>Initiation</i>    |
|--------------------------------------------|--------------------|---------------------|--------------------|----------------------|
| Sample                                     | Payers<br>(1)      | Payers<br>(2)       | Non-Payers<br>(3)  | Non-Payers<br>(4)    |
| $Treat \times Post$                        | 0.0126<br>(0.0167) |                     | 0.0168<br>(0.0122) |                      |
| $Treat \times Pre_{2015}$                  |                    | -0.0013<br>(0.0257) |                    | 0.0273<br>(0.0183)   |
| $Treat \times Post_{2016}$                 |                    | 0.0505*<br>(0.0269) |                    | 0.0313<br>(0.0208)   |
| $Treat \times Post_{2017}$                 |                    | 0.0030<br>(0.0270)  |                    | 0.0516**<br>(0.0207) |
| $Treat \times Post_{2018}$                 |                    | -0.0225<br>(0.0267) |                    | 0.0240<br>(0.0230)   |
| Observations                               | 26,413             | 26,413              | 10,232             | 10,232               |
| Adjusted R-squared                         | 0.1686             | 0.1688              | -0.0063            | -0.0063              |
| Pre-event controls $\times$ Year           | Y                  | Y                   | Y                  | Y                    |
| Firm FE                                    | Y                  | Y                   | Y                  | Y                    |
| Country $\times$ Industry $\times$ Year FE | Y                  | Y                   | Y                  | Y                    |